

Negative Nominal Interest Rates and Monetary Policy*

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Abstract

How much can central banks reduce nominal interest rates? Can the lower bound be controlled by monetary policy? If so, should central banks reduce it to implement negative interest rates? I construct a model with multiple means of payment where the costs of holding paper currency effectively reduce its rate of return, creating a negative effective lower bound on interest rates. I find that central banks can reduce this lower bound with *a non-par exchange rate* between currency and bank reserves, but doing so raises currency-holding costs for individuals, leading to welfare losses. Moreover, implementing a negative rate by reducing the lower bound has no benefits because this policy combination lowers both the rate of return on currency and the interest rate on financial assets, leaving relative interest rates unchanged.

Keywords: Negative interest rate; Effective lower bound; Money; Banking; Monetary policy

JEL Codes: E4; E5

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1 Introduction

Since the global financial crisis, negative nominal interest rates have become a part of the monetary policy toolkit for some prominent central banks in the world such as the European Central Bank, the Swiss National Bank, the Swedish Riksbank, the Bank of Japan, and the National Bank of Denmark. Although negative nominal interest rates are feasible, the lower bound on nominal interest rates can still be an important binding constraint on monetary policy.¹ Accordingly, there has been much discussion about policy tools aimed at reducing this lower bound (see, for example, [Goodfriend, 2016](#); [Rogoff, 2017a,b](#); [Agarwal and Kimball, 2015, 2019](#)).

Understanding how and why central banks could control the lower bound requires uncovering its determinants. What economic fundamentals or frictions determine the lower bound? Would it be desirable to manipulate these frictions to reduce the lower bound? I answer these questions by developing a model where the lower bound on nominal interest rates is endogenously negative and responds to central bank policies. My main finding is that while central banks can reduce this lower bound, such efforts only result in welfare losses.

In standard macroeconomic theory, monetary policy is constrained by the zero lower bound on nominal interest rates.² This constraint arises because of arbitrage: borrowing at a negative rate and investing in zero-interest paper currency would be profitable if nominal interest rates were negative. However, as noted above, negative nominal short-term interest rates are implementable in practice. This suggests the presence of frictions that inhibit arbitrage, making the lower bound on nominal interest rates effectively negative.

The frictions that create a negative effective lower bound are inherent to paper currency and emerge from the costs associated with storing, transporting, and exchanging it in large quantities. For example, if a private bank faces a negative nominal interest rate on reserves, it might contemplate holding currency instead of having reserve balances with the central bank. However, this would entail costs from installing a sizable vault and hiring security guards to watch it. Moreover, the currency held would be of little or no use in making interbank and online transactions.

The policy tools aimed at reducing the effective lower bound essentially work by enhancing or taking advantage of the frictions to reduce the effective rate of return on holding currency.³ A lower return on currency leads to a lower effective lower bound, which permits potentially welfare-

¹The effective lower bound was a binding constraint for some central banks during the 2010s, as they have been cautious about implementing negative or substantially negative interest rates. Although the Swiss National Bank and the National Bank of Denmark successfully set a record-low policy rate of -0.75 percent, it remains uncertain whether they could have further reduced their policy rates and whether other central banks could set their policy rates at that level. [Witmer and Yang \(2016\)](#) find that the effective lower bound for the Bank of Canada policy rate ranges between -0.25 percent and -0.75 percent.

²See, for example, [Woodford \(2003\)](#) and [Curdia and Woodford \(2010\)](#) for standard New Keynesian models. See, also, [Lagos and Wright \(2005\)](#) and [Lagos, Rocheteau, and Wright \(2017\)](#) for standard New Monetarist models.

³Some policy tools directly affect the currency-holding costs mentioned earlier. Imposing a quantitative limit on cash withdrawals from the central bank cash window or eliminating large-denomination bills increases the storage and/or transportation costs of currency. Other policy tools, including a non-par exchange rate between currency and reserves that I study in this paper, are intended to indirectly increase the cost of holding currency by reducing its nominal rate of return.

enhancing monetary policy. However, reducing the currency’s effective rate of return also implies an increased cost of using currency as a means of payment and a potential welfare loss. I formally examine this potential tradeoff and evaluate the welfare implications of reducing the effective lower bound on nominal interest rates.

Specifically, I develop a model with two means of payment—currency and bank deposits—and introduce currency-holding costs that lead to a negative effective lower bound. Also, to demonstrate how a central bank can reduce this effective lower bound, I consider a market-based reserve policy. This policy involves altering the one-to-one exchange rate between currency and bank reserves, as proposed by [Eisler \(1932\)](#), [Buiter \(2010\)](#) and [Agarwal and Kimball \(2015\)](#).⁴ As the central bank can set a different exchange rate for private banks’ current currency withdrawals from the one for their future currency deposits, it can reduce the nominal rate of return on currency faced by private banks.

The effectiveness of this reserve policy—a *non-par* exchange rate for currency withdrawals—crucially depends on its ability to reduce the effective rate of return on currency faced by private individuals. I show that the reserve policy is effective because of currency-holding costs, which incentivize individuals to continue depositing and withdrawing currency even though it yields a negative return. That is, the central bank can successfully lower the effective lower bound by taking advantage of the frictions related to currency.

The ability of a non-par exchange rate to reduce the effective lower bound, however, can be limited. If individuals choose not to deposit their currency at banks, the central bank loses its ability to reduce the effective lower bound. This can happen if a non-par exchange rate creates incentives for currency side-trading in the private sector. Specifically, private banks are willing to acquire currency at a price (in units of reserves) lower than the non-par exchange rate of reserves for currency withdrawals, while private individuals are willing to unload their currency at a price higher than one unit of reserves. When the marginal cost of holding currency does not increase with currency holdings, individuals deposit currency at banks only when the marginal benefit from side-trading is sufficiently low, i.e., when the non-par exchange rate for currency withdrawals is sufficiently low.

This result can be reversed if the marginal cost of holding currency increases with currency holdings. This can happen endogenously when a larger amount of currency held by individuals promotes currency theft. That is, the cost of holding currency can increase at the margin either because theft occurs more frequently or because individuals must incur higher security costs to prevent it. I model this by allowing individuals to steal currency, at a cost, from those engaged in side-trades, so that theft arises endogenously.⁵ The resulting increase in the cost of holding currency encourages individuals to deposit their currency at banks. Consequently, the central bank

⁴In particular, [Agarwal and Kimball \(2019\)](#) consider this unconventional reserve policy as “the first-best approach with the fewest undesirable side-effects” because the nominal rate of return on currency in units of reserves, created by the central bank, can be naturally transmitted to the rate of return on currency in units of other financial assets.

⁵An endogenous cost of holding currency in a form of endogenous theft is also modeled in [He, Huang, and Wright \(2005, 2008\)](#) and [Sanches and Williamson \(2010\)](#).

can always reduce the effective lower bound on nominal interest rates.

I then turn to the consequences of reducing the effective lower bound and implementing a lower nominal interest rate on reserves. Currency-holding costs create inefficiencies in ordinary transactions by reducing the quantity of currency, in real terms, used as a means of payment, thereby decreasing production and consumption. Lowering the nominal interest rate can improve welfare by effectively increasing the real return on currency. This encourages individuals to use more currency as a means of payment, thus mitigating the inefficiencies in ordinary transactions.⁶ However, reducing the effective lower bound encourages more individuals to engage in socially useless and costly currency side-trading and theft, decreasing welfare.

Despite the apparent tradeoff, I find that it is never optimal to reduce the effective lower bound, as doing so only increases the cost of holding currency without yielding any welfare gains. Intuitively, lowering the nominal interest rate on reserves encourages private banks to hold more currency relative to interest-bearing assets. However, reducing the effective lower bound, which involves lowering the nominal rate of return on currency, does the opposite: it incentivizes private banks to hold less currency relative to interest-bearing assets. So, if the central bank reduces both the interest rate and the lower bound by the same magnitude, the effect of reducing the lower bound completely offsets that of reducing the interest rate. In other words, it is the interest rate on reserves relative to currency that determines the monetary policy stance. Because the central bank cannot further reduce this relative interest rate, there are no gains from reducing the effective lower bound.

Finally, I explore how disintermediation can affect the effective lower bound on nominal interest rates. Disintermediation happens when a sufficiently low nominal interest rate makes more individuals opt to use currency rather than bank deposits to make ordinary transactions. This is a practical concern because bank deposits serve as a primary and stable funding source for financing bank loans, and thus, disintermediation could lead to long-run inefficiency in the financial system. To study the role of disintermediation, I extend my model to allow individuals to opt out of the banking system.

I find that a non-par exchange rate policy enables central banks to set a negative nominal interest rate without causing disintermediation. However, this reserve policy also encourages costly theft and results in welfare losses, as in the baseline model. In an extreme case where individuals can make all transactions with currency, the nominal interest rate must be sufficiently high to support banking activities. I show that the lower bound on nominal interest rates preventing complete disintermediation can be higher than the one preventing arbitrage. This implies that the effective lower bound is determined by the lower bound on disintermediation-free interest rates.⁷

⁶This is a standard property in monetary theory with implications for optimal monetary policy. Similar to [He, Huang, and Wright \(2008\)](#), I show that the presence of currency-holding costs can make the optimal nominal interest rate negative. See [Lagos, Rocheteau, and Wright \(2017\)](#) for more discussion about optimal monetary policy.

⁷My analysis abstracts from the conflict between the two distinct lower bounds, which is a driving force for the imperfect pass-through of negative policy rates in [Eggertsson, Juelsrud, Summers, and Wold \(2022\)](#). However, my findings suggest that a non-par exchange rate policy can reduce both lower bounds by increasing currency-holding costs faced by individuals.

Literature Review This paper contributes to the literature on how to reduce the effective lower bound on nominal interest rates. [Agarwal and Kimball \(2019\)](#) provide a comprehensive survey and discuss the pros and cons of policy tools suggested in the literature. The idea of a non-par exchange rate between currency and reserves was first proposed by [Eisler \(1932\)](#) in the form of a dual currency system where one currency (physical currency) is used as a means of payment, and the other (electronic money) plays a unit-of-account role. [Buiter \(2010\)](#) revived Eisler’s proposal with a simple model illustrating how the central bank could frictionlessly reduce the lower bound by adjusting the exchange rate between the two currencies. More recently, [Agarwal and Kimball \(2015\)](#), [Goodfriend \(2016\)](#), and [Rogoff \(2017a,b\)](#) have favorably discussed a non-par exchange rate between currency and reserves as a potential policy tool. I formalize the insights presented in the literature and examine the frictions that determine the effective lower bound. Furthermore, I demonstrate how a non-par exchange rate can effectively reduce the rates of return on currency faced by private banks and individuals.

This paper is also related to theoretical studies examining the implications of negative nominal interest rates, including [He, Huang, and Wright \(2008\)](#), [Rognlie \(2016\)](#), [Jung \(2019\)](#), [Ulate \(2021\)](#), [Eggertsson, Juelsrud, Summers, and Wold \(2022\)](#), and [Abadi, Brunnermeier, and Koby \(2023\)](#). [He, Huang, and Wright \(2008\)](#) develop a model of currency and bank deposits, similar to mine, where using currency is relatively less safe due to the risk of theft. They show that a negative nominal interest rate can be optimal under certain conditions. My paper departs from theirs by distinguishing between private banks’ deposits with the central bank (reserves) and private individuals’ bank deposits. This enables an examination of how a non-par exchange rate between currency and reserves impacts the actual rate of return on currency and the terms of bank deposit contracts. Another related paper is [Eggertsson, Juelsrud, Summers, and Wold \(2022\)](#), which shows that the lower bound on disintermediation-free interest rates (deposit rates) can exceed the lower bound on arbitrage-free policy rates (short-term interest rates). They find that when the former lower bound is binding while the latter is not, implementing a negative nominal interest rate policy can lead to contractionary effects due to the breakdown in the pass-through of the policy rate. In contrast, my paper focuses specifically on examining how a central bank can reduce the lower bound on nominal interest rates and its welfare implications.

2 Baseline Model

The basic structure of the model is similar to that of [Lagos and Wright \(2005\)](#) and [Rocheteau and Wright \(2005\)](#). Time is indexed by $t = 0, 1, 2, \dots$, and there are three subperiods in each period. The theft market (TM) opens in the first subperiod, the centralized market (CM) opens in the following subperiod, and the decentralized market (DM) opens in the last subperiod. I incorporate the theft market into the standard framework to reflect frictions limiting the size of currency side-trades, as

elaborated later. There is a continuum of buyers with unit mass, each of whom maximizes

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[-\tilde{H}_t - H_t + u(x_t) \right], \quad (1)$$

where $0 < \beta < 1$. Here, $\tilde{H}_t$ and H_t denote the buyer's labor supply in the TM and the CM, respectively, and x_t denotes their consumption in the DM. The utility function $u(\cdot)$ is assumed to be strictly increasing, strictly concave, and twice continuously differentiable, with $u'(0) = \infty$, $u'(\infty) = 0$, and $-\frac{xu''(x)}{u'(x)} < 1$. There is also a continuum of sellers with unit mass, each of whom maximizes

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[-\tilde{H}_t^s + X_t^s - h_t^s \right]. \quad (2)$$

Here, X_t^s represents the seller's consumption in the CM, and $\tilde{H}_t^s$ and h_t^s denote their labor supply in the TM and the DM, respectively. Finally, there is a continuum of private banks. Private banks are agents who are active only in the CM, each of whom maximizes

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[X_t^b - H_t^b \right]. \quad (3)$$

Here, X_t^b represents the bank's consumption in the CM, and H_t^b represents their labor supply in the CM. In the CM or the DM, one unit of perishable consumption good can be produced with one unit of labor supply, while no production of goods occurs in the TM. Buyers cannot produce goods in the DM, while sellers cannot produce in the CM.

At the beginning of the TM, sellers holding currency have the option to safely deposit their currency with the central bank and receive reserve balances in exchange.⁸ At this stage, one unit of currency is exchanged for one unit of reserve balances, which sellers can exchange for goods in the following CM. After sellers make currency deposits with the central bank, buyers can invest κ units of labor to acquire a theft technology (e.g., producing a weapon). Buyers and sellers are then randomly matched. In cases where a buyer equipped with the theft technology meets a seller carrying currency, the buyer can steal all of the seller's currency.

At the beginning of the CM, debts are paid off, then production, consumption, and asset exchanges take place in a perfectly competitive market. Private banks can obtain currency in three different ways—from a seller, from a buyer (stolen currency), or by acquiring reserves with the central bank and exchanging the reserves for currency. Also, private banks write deposit contracts with buyers before buyers learn their types. Here, a buyer's type refers to the type of seller they will meet in the following DM, as specified in what follows. These deposit contracts provide insurance to

⁸It is worth noting that, in practice, private individuals typically cannot hold reserve accounts directly with the central bank. The assumption made here could be interpreted as individuals depositing currency with a private bank, which then exchanges the currency for reserve balances. Alternatively, this assumption could be interpreted as a scenario where an individual deposits currency with the central bank, and the central bank credits the corresponding private bank, which, in turn, credits the individual's bank account.

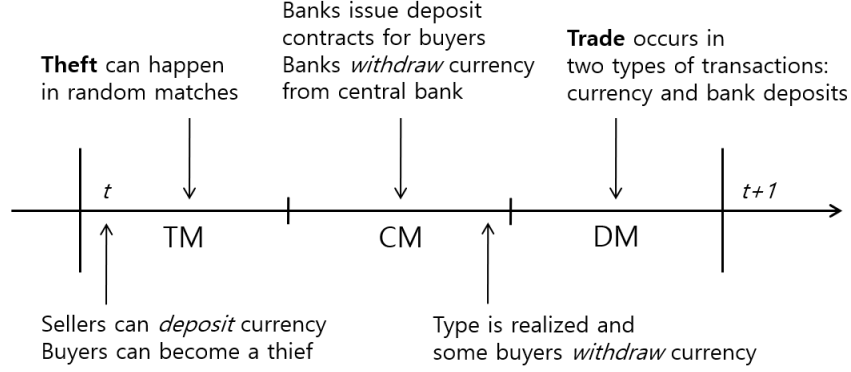


Figure 1: Timing of events

buyers, by allowing them to withdraw currency at the end of the CM when they learn their types. Buyers' types are publicly observable.

In the DM, each buyer is randomly matched with a seller and makes a take-it-or-leave-it offer to the seller. In any DM matches, a matched buyer and seller do not know each others' histories (no memory or record keeping) and are subject to limited commitment. This implies that buyers' IOUs cannot be traded in the DM. There are two types of sellers. A fraction ρ of sellers accepts only currency, while the remaining fraction $1 - \rho$ accepts only claims on banks. The timing of events is summarized in Figure 1.

Some sellers who obtain currency in the DM may choose to exchange it for goods in the following CM rather than safely depositing it with the central bank. Let α_t^s denote the fraction of sellers who bring currency into the CM among those who obtained it in the previous DM. Also, let α_t^b denote the fraction of buyers who invest in the theft technology. Then, the probability that each seller meets a buyer who steals currency is α_t^b , while the probability that each buyer meets a seller holding currency is $\rho\alpha_t^s$.

The fundamental assets in this economy are currency, reserves, and nominal government bonds. Currency and reserves are issued by the central bank. Currency is perfectly divisible, portable, and storable, and bears a nominal interest rate of zero. Reserves are private banks' account balances with the central bank, and one unit of reserves acquired in the CM of period t pays off R_{t+1}^m units of reserves at the beginning of the CM in period $t + 1$. Private banks can withdraw currency from their reserve accounts by visiting the central bank. Following the ideas presented in Eisler (1932), Buiter (2010) and Agarwal and Kimball (2015), the central bank can set a *non-par* exchange rate between currency and reserves to create a negative nominal rate of return on currency. That is, depositing one unit of currency with the central bank yields one unit of reserves, while withdrawing one unit of currency in period t costs $\eta_t \geq 1$ units of reserves. Nominal government bonds, issued by the fiscal authority, are one-period bonds which pay off a gross nominal interest rate of R_t^b .

In addition to the fundamental assets supplied by the consolidated government, private banks will endogenously create private assets in the form of deposit claims. I assume that private banks

have a collateral technology that allows creditors to seize at least part of the asset in case of default. As deposit claims can be secured by collateral (banks' assets), they can serve as means of payment in the DM. Private banks can hold currency in their asset portfolio from the CM of period t until the next CM. However, this incurs a storage cost γc_t in the next CM, where $0 < \gamma < 1$ and c_t is the quantity of currency in real terms acquired in the CM of period t . The proportional cost of storing currency across periods has implications for the lower bound on nominal interest rates, as will be shown later. Sellers have the same storage technology as do banks, but buyers cannot hold currency across periods.⁹ Also, individuals holding currency at the end of the DM incur a fixed cost μ . Eventually, sellers who receive currency in the DM will bear this cost, which will generate inefficiency in transactions involving currency.¹⁰

2.1 Government

Confine attention to stationary equilibria where all real variables and government policies are constant across periods. I assume that the consolidated government starts issuing its liabilities with no outstanding debt in period 0. The consolidated government's budget constraint at $t = 0$ is expressed as:

$$\eta \bar{c} + \bar{m} + \bar{b} = \tau_0. \quad (4)$$

Here, $\bar{c}$, $\bar{m}$, and $\bar{b}$ denote the quantities of currency, reserves, and nominal government bonds outstanding, in real terms, at the end of period 0 (and in every following period). Additionally, τ_0 denotes the real quantity of lump-sum transfers to each buyer. The fiscal authority can levy lump-sum taxes on buyers in equal amounts. So, the consolidated government's budget constraint at $t = 1, 2, \dots$ can be expressed as:

$$\eta \bar{c} + \bar{m} + \bar{b} = \frac{\bar{c} + R^m \bar{m} + R^b \bar{b}}{\pi} + \tau, \quad (5)$$

where π is the gross inflation rate, and τ is the real quantity of lump-sum transfers (or lump-sum taxes if $\tau < 0$) to each buyer. The left-hand side of equation (5) represents the total revenue of the consolidated government from issuing new liabilities, while the right-hand side represents its total expenditure—the sum of the debt repayments and the transfers to buyers.

As in [Andolfatto and Williamson \(2015\)](#) and [Williamson \(2016\)](#), I assume that the fiscal authority determines the real value of the consolidated government debt outstanding, denoted as v ,

⁹To make the model analytically tractable, I assume that buyers cannot store currency across periods. Relaxing this assumption would only complicate the model without adding any useful insights.

¹⁰Introducing a fixed cost is sufficient to generate inefficiency in currency-involved transactions, which has implications for optimal monetary policy. Adding a proportional cost for currency holders at the end of the DM would make the model analytically intractable.

where

$$v = \eta \bar{c} + \bar{m} + \bar{b}, \quad (6)$$

for all periods.¹¹ This implies that, from equations (4) and (5), the fiscal authority adjusts the quantities of lump-sum transfers τ_0 and τ to achieve its policy goal. Given this fiscal policy that targets the total value of the consolidated government debt, the central bank's monetary policy serves to alter its composition. This fiscal policy rule is a useful specification that creates a low real interest rate in the economy. As will be shown later, a low value of consolidated government debt (or a low v) translates into a low supply of collateralizable assets, thereby resulting in binding collateral constraints, a liquidity premium on those assets, and consequently, low real interest rates.

3 Equilibrium

In this section, I will describe how buyers and sellers make their decisions in the TM and how private banks choose their deposit contracts and asset portfolios in the CM. Then, I will define and characterize an equilibrium and show why the effective lower bound on nominal interest rates can be negative.

3.1 Currency Side-Trading and Theft

In the CM, private banks can withdraw currency from the central bank by paying η units of their reserve balances per unit of currency. If there are would-be sellers of currency in the CM, private banks would be willing to participate in currency side-trades to reduce the cost of acquiring currency. In equilibrium, private banks must be indifferent between withdrawing currency from the central bank and obtaining it from any would-be sellers of currency. This implies that the price of currency in units of reserves in the CM must be η in equilibrium.¹²

Currency side-trading between private banks and currency holders can occur in the CM only when there are sellers who chose not to deposit their currency in the previous TM. However, in

¹¹In period 0, the central bank purchases government bonds by issuing currency and reserves, and it transfers its profits to the fiscal authority in every following period. This implies that the real value of the central bank's assets must be equal to that of its liabilities in every period, that is,

$$\eta \bar{c} + \bar{m} = \hat{b},$$

where $\hat{b}$ denotes the real quantity of government bonds held by the central bank. Therefore, the fiscal authority effectively determines the real quantity of total government bonds it issues because

$$v = \hat{b} + \bar{b}.$$

¹²This scenario occurs when the real quantity of currency traded in the CM does not exceed the real quantity demanded by private banks. Specifically, I focus on stationary equilibria with sufficiently low values of v and γ , ensuring that there is inflation $\pi > 1$ in equilibrium. Due to inflation, the real quantity of currency traded consistently falls short of the real quantity demanded. Therefore, private banks must withdraw some currency from the central bank every period. See Appendix A.1 for the details on the conditions that lead to stationary equilibria with deflation.

the TM, some buyers may invest κ units of labor supply to acquire the theft technology. Suppose that the representative currency-holding seller carries c^s units of currency, in real terms, into the TM. As acquiring the theft technology is costly, each buyer's decision on stealing currency must be incentive-compatible in equilibrium. That is,

$$\text{if } \kappa > \rho\alpha^s\eta c^s, \quad \text{then } \alpha^b = 0, \quad (7)$$

$$\text{if } \kappa = \rho\alpha^s\eta c^s, \quad \text{then } 0 \leq \alpha^b \leq 1, \quad (8)$$

$$\text{if } \kappa < \rho\alpha^s\eta c^s, \quad \text{then } \alpha^b = 1, \quad (9)$$

where $\rho\alpha^s$ is the probability of meeting a currency-holding seller, and ηc^s is the real value of currency held by the seller. Conditions (7)-(9) state that theft does not occur if the buyer strictly prefers not to steal currency, the buyer sometimes steals if they are indifferent between the two options, and the buyer always steals if theft is strictly preferred. Similarly, each seller's decision on carrying currency into the TM must be incentive-compatible, that is,

$$\text{if } (1 - \alpha^b)\eta < 1, \quad \text{then } \alpha^s = 0, \quad (10)$$

$$\text{if } (1 - \alpha^b)\eta = 1, \quad \text{then } 0 \leq \alpha^s \leq 1, \quad (11)$$

$$\text{if } (1 - \alpha^b)\eta > 1, \quad \text{then } \alpha^s = 1. \quad (12)$$

The seller does not carry the currency into the TM if the expected payoff from carrying one unit of currency $(1 - \alpha^b)\eta$ is less than the payoff from depositing it with the central bank and obtaining one unit of reserves. If the seller is indifferent between the two choices, they sometimes carry the currency into the TM. Otherwise, the seller always carries the currency into the TM.

3.2 Deposit Contracts

Private banks write deposit contracts for buyers in the CM before buyers learn their types. These deposit contracts provide insurance to buyers, as in [Williamson \(2016, 2022\)](#), by giving them an option to withdraw currency after they learn their types. Buyers who do not exercise this option are those who can use deposit claims as a means of payment in the DM. Suppose a bank proposes a deposit contract (k, c', d) , where k is the quantity of CM goods deposited by each buyer at the beginning of the CM, c' is the real quantity of currency that the buyer can withdraw at the end of the CM, and d is the quantity of claims to CM goods in the following period that the buyer can exchange in the DM if currency has not been withdrawn. Also, the bank acquires an asset portfolio (b, m, c) , where b is the quantity of government bonds, m is the quantity of reserves, and c is the quantity of currency in real terms. In equilibrium, the bank's optimization problem can be written as:

$$\max_{k, c', d, b, m, c} \left\{ -k + \rho u \left([1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta] \frac{\beta c'}{\pi} - \beta \mu \right) + (1 - \rho)u(\beta d) \right\} \quad (13)$$

subject to

$$k - b - m - \eta c + \beta \left[-(1 - \rho)d + \frac{R^m m + R^b b + c - \rho c'}{\pi} - \gamma(c - \rho c') \right] \geq 0, \quad (14)$$

$$-(1 - \rho)d + \frac{R^m m + R^b b + c - \rho c'}{\pi} \geq \frac{\delta (R^m m + R^b b + c)}{\pi}, \quad (15)$$

$$k, c', d, b, m, c, c - \rho c' \geq 0. \quad (16)$$

The objective function (13) is the representative buyer's expected utility, indicating that the bank chooses a contract that maximizes the buyer's expected utility in equilibrium. With probability ρ , the buyer realizes that, in the following DM, they will be matched with a seller who accepts only currency. In this case, the buyer visits the bank to withdraw c' units of currency at the end of the CM. In the following DM, the buyer makes a take-it-or-leave-it offer to the matched seller, acquiring $[1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta] \frac{\beta c'}{\pi} - \beta \mu$ units of goods.¹³ With probability $1 - \rho$, the buyer learns that they will meet a seller who accepts claims on the bank. As the buyer does not withdraw currency in this case, they receive a claim to d units of goods in the next CM. So, the buyer's take-it-or-leave-it offer implies that the buyer trades d deposit claims for βd units of goods.

Constraint (14) states that the bank earns a nonnegative discounted net payoff in equilibrium. In the CM, the bank receives k deposits from the buyer and acquires a portfolio of government bonds b , reserves m , and currency c . At the end of the CM, the bank pays out currency to the fraction ρ of buyers, each of whom withdraws c' currency. The remaining fraction $1 - \rho$ of buyers receive deposit claims, and in the following CM, the bank pays off d units of goods to each deposit claim holder. Notice that the bank stores the remaining $c - \rho c'$ units of currency until the next CM, which incurs a storage cost of $\gamma(c - \rho c')$ units of labor supply. At the beginning of the next CM, the bank must deposit the remaining currency with the central bank at a one-to-one exchange rate.

Like any other agents in the economy, the bank is subject to limited commitment. This implies that the bank's deposit liabilities must be backed by collateral, and (15) represents a collateral constraint. The bank can abscond with a fraction δ of its assets, pledged as collateral, when it defaults. So, the collateral constraint tells us that the bank must weakly prefer to repay its deposit liabilities in the CM and in the next CM rather than absconding with collateral. If the bank were to default, it would not allow buyers to withdraw currency. Finally, constraint (16) demonstrates that all real quantities must be nonnegative.

¹³Given that the buyer makes a take-it-or-leave-it offer to the seller, they can extract all surplus from trade. By accepting the buyer's offer, the seller receives c' units of currency (in real terms) in the DM. Then, in the next period, the seller will hold $\frac{c'}{\pi}$ units of currency and bear μ units of fixed cost (a disutility from supplying labor) at the beginning of the TM. The ex-ante expected payoff per unit of currency is $1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta$, because with probability $1 - \alpha^s$, the seller deposits the currency to receive one unit of reserves, and with probability $\alpha^s(1 - \alpha^b)$, the seller successfully sells the currency in the CM at price η . As the seller's surplus from trade becomes zero, the quantity of goods produced by the seller and transferred to the buyer (or equivalently, the disutility from producing goods) equals the seller's discounted expected net payoff from acquiring c' units of currency in the DM.

3.3 Definition of Equilibrium

Any contract that provides a positive discounted net payoff for the bank cannot be supported in equilibrium. If private banks were to earn a positive discounted net payoff, a bank would design an alternative contract that provides a slightly lower payoff per contract but a higher total payoff by attracting all buyers. So, constraint (14) must hold with equality in equilibrium.

Let λ denote the Lagrange multiplier associated with the collateral constraint (15). Then, I can derive the first-order conditions for the bank's optimization problem, (13) subject to (14)-(16), as follows:

$$\frac{\beta[1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta]}{\pi} u' \left([1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta] \frac{\beta c'}{\pi} - \beta\mu \right) - \eta - \frac{\lambda\delta}{\pi} = 0, \quad (17)$$

$$\beta u'(\beta d) - \beta - \lambda = 0, \quad (18)$$

$$-1 + \frac{\beta R^m}{\pi} + \frac{\lambda R^m(1 - \delta)}{\pi} = 0, \quad (19)$$

$$-1 + \frac{\beta R^b}{\pi} + \frac{\lambda R^b(1 - \delta)}{\pi} = 0, \quad (20)$$

$$-\eta + \frac{\beta}{\pi} - \beta\gamma + \frac{\lambda(1 - \delta)}{\pi} \leq 0, \quad (21)$$

$$\lambda \left[-(1 - \rho)d + \frac{(1 - \delta)(R^m m + R^b b + c)}{\pi} - \frac{\rho c'}{\pi} \right] = 0. \quad (22)$$

A necessary condition for an equilibrium to exist is that sellers do not hold currency from the CM to the next CM. So, the expected payoff from holding currency across periods must be nonpositive at the margin. That is,

$$-\eta + \beta \left[\frac{1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta}{\pi} - \gamma \right] \leq 0. \quad (23)$$

Also, in equilibrium, asset markets clear in that the demand for each asset is equal to the supply. That is,

$$c = \bar{c}; \quad m = \bar{m}; \quad b = \bar{b}. \quad (24)$$

For convenience, let x^c and x^d denote the consumption quantities in DM meetings, respectively, with currency and deposit claims being traded, i.e.,

$$x^c = [1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta] \frac{\beta c'}{\pi} - \beta\mu, \quad (25)$$

$$x^d = \beta d. \quad (26)$$

I assume that the central bank conducts monetary policy under a floor system where a sufficiently large quantity of reserve balances are held by private banks, and the central bank sets the nominal interest rate on reserves R^m . Under this system, private banks treat reserves and government bonds

as identical assets at the margin, so the nominal interest rate on reserves pegs the nominal interest rate on government bonds in equilibrium, i.e., $R^m = R^b$ from (19) and (20).¹⁴ Also, the central bank can expand its balance sheet through swaps of reserves for government bonds. Let $\omega = \eta\bar{c} + \bar{m}$ denote the size of the balance sheet, which is equivalent to the real value of the central bank's liabilities.¹⁵

Then, I can define an equilibrium as follows.

Definition *Given exogenous fiscal policy v and monetary policy (R^m, ω, η) , a stationary equilibrium consists of DM consumption quantities (x^c, x^d) , asset quantities (k, c', d, b, m, c) , the fraction of buyers who choose to steal currency in the TM α^b , the fraction of sellers who choose to carry currency into the TM among those who acquired it in the previous DM α^s , transfers (τ_0, τ) , the gross inflation rate π , and the gross nominal interest rate on government bonds R^b , satisfying the consolidated government budget constraints (4) and (5), the fiscal policy rule (6), the first-order conditions for the bank's problem (17)-(22), the no-arbitrage condition for sellers (23), the incentive compatibility conditions for buyers and sellers (7)-(12), and market clearing conditions (24).*

Notice that, according to the definition, the fiscal and monetary policies are exogenous. While the fiscal authority determines the total value of consolidated government debt, in real terms, the central bank has three policy targets: (i) the nominal interest rate on reserves R^m , (ii) the size of the central bank's balance sheet ω , and (iii) the non-par exchange rate for currency withdrawals η . Also, note that the fiscal authority adjusts transfers τ_0 and τ , in response to monetary policy, so as to satisfy its fiscal policy target v and the government's budget constraints.

3.4 Characterization of Equilibrium

First, I will characterize the effective lower bound (ELB) on the gross nominal interest rate R^m . The ELB will be defined as a critical threshold that ensures no arbitrage opportunities for private banks and sellers from holding currency across periods. The inequality (21) ensures that no arbitrage opportunities exist for private banks from acquiring currency in the CM, holding it across periods, and depositing it with the central bank in the next CM. Substituting (19) into (21) gives:

$$R^m \geq \frac{1}{\eta + \beta\gamma}.$$

¹⁴In practice, reserves are considered as a useful means of payment in intraday trading in the banking system. However, a key property of the U.S. financial system in the post-financial crisis period is that a large volume of reserves has been held by private banks without being used in intraday financial transactions. This observation allows us to abstract from the role of reserves in intraday transactions. Although only reserves can be turned into currency through the central bank cash window, this does not create an interest rate differential between reserves and government bonds.

¹⁵The real value of liabilities is equal to that of assets because the central bank is assumed to transfer any profits/losses to the fiscal authority, and the central bank's net worth is zero.

Similarly, using (19), the no-arbitrage condition for sellers from holding currency across periods (inequality 23) can be written as:

$$R^m \geq \frac{1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta}{(\eta + \beta\gamma)[(1 - \delta)u'(x^d) + \delta]}.$$

The gross nominal interest rate R^m must be sufficiently high to prevent arbitrage for both private banks and sellers. Thus, the ELB on the gross nominal interest rate in equilibrium can be defined as follows:

$$R^m \geq \max \left\{ \frac{1}{\eta + \beta\gamma}, \frac{1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta}{(\eta + \beta\gamma)[(1 - \delta)u'(x^d) + \delta]} \right\} \equiv ELB. \quad (27)$$

To understand why the gross nominal interest rate R^m can be less than one (or the net nominal interest rate $R^m - 1$ can be negative), suppose first that the ELB on R^m is determined by the first argument. Then, the ELB can be less than one for two reasons. First, the proportional cost associated with holding currency across periods, represented by $\gamma > 0$, causes the ELB to fall below one. This result can explain why some central banks were able to implement negative nominal interest rates without causing immediate arbitrage or disruptions in the financial system. Also, a non-par exchange rate for currency withdrawals, represented by $\eta > 1$, drives the ELB to fall below zero. This occurs because a higher exchange rate for currency withdrawals than that for currency deposits decreases the nominal rate of return on holding currency across periods.

Now, suppose that the ELB on R^m is determined by the second argument in (27). Then, there arises a nonstandard reason for the ELB to be less than one. As I will show later, the term $(1 - \delta)u'(x^d) + \delta$ is greater than one due to low real interest rates on collateralizable assets. The real interest rates are low because assets that are useful as collateral bear a liquidity premium, driving up asset prices. However, sellers do not receive non-monetary payoffs from using currency as collateral, and thus, the rate of return on currency perceived by sellers is lower than that perceived by banks. When the liquidity premium is sufficiently high or the real interest rate is sufficiently low, the ELB can be lower than one. For analytical convenience, I will focus on cases where inequality (27) holds with strict inequality, meaning that the nominal interest rate on reserves R^m is not constrained by the ELB.

Suppose each seller who received currency in the previous DM holds c^s units of currency at the beginning of the TM. Then, the following lemma provides insights into the equilibrium behavior of buyers and sellers in the TM, represented by α^b and α^s .

Lemma 1 *Suppose that $\eta = 1$. Then, no theft occurs in equilibrium, i.e., $\alpha^b = 0$. Furthermore, $\alpha^s \in [0, 1]$ for all $\kappa \geq \rho\eta c^s$, and $\alpha^s \in [0, \bar{\alpha}^s]$ for all $\kappa < \rho\eta c^s$ where $\bar{\alpha}^s = \frac{\kappa}{\rho\eta c^s}$. Alternatively, suppose that $\eta > 1$. Then, no theft occurs in equilibrium with $\alpha^b = 0$ and $\alpha^s = 1$ for all $\kappa \geq \rho\eta c^s$, while theft exists in equilibrium with $\alpha^b = \frac{\eta-1}{\eta}$ and $\alpha^s = \frac{\kappa}{\rho\eta c^s}$ for all $\kappa < \rho\eta c^s$.*

Proof See Appendix $\square$

According to Lemma 1, theft does not occur in equilibrium when private banks can withdraw currency from their reserve accounts at par. In this case, the price of currency is identical to that of reserves in terms of goods, making sellers indifferent between depositing the currency with the central bank and engaging in side trades with private banks. Since sellers would opt to deposit their currency until theft disappears, there is no incentive for costly theft. Another finding is that no theft occurs when the exchange rate for currency withdrawals is non-par ($\eta > 1$), but the cost of theft κ is too high. In this case, all sellers prefer to carry their currency rather than depositing it with the central bank, i.e., $\alpha^s = 1$. Finally, when the cost of theft is sufficiently low, a non-par exchange rate for currency withdrawals encourages sellers to participate in currency side-trading. With more sellers carrying currency, the marginal benefit for buyers investing in the theft technology increases. Conversely, with more buyers having the theft technology, the marginal cost for sellers carrying currency rises. In equilibrium, both buyers and sellers become indifferent between their choices.

From (25) and Lemma 1, I can show that:

$$x^c = \frac{\beta c' \eta}{\pi} - \beta \mu, \quad \forall \kappa \geq \frac{\rho \eta c'}{\pi} \quad (28)$$

$$x^c = \frac{\beta c'}{\pi} - \beta \mu. \quad \forall \kappa < \frac{\rho \eta c'}{\pi} \quad (29)$$

In what follows, I will consider cases where the real value of the consolidated government debt outstanding v is sufficiently low to ensure that collateral constraints are binding in equilibrium.

3.5 Equilibrium with No Theft

In this section, I analyze an equilibrium where all currency-holding sellers engage in currency side-trading with private banks in the CM, and there is no threat of theft in the TM. Suppose that the cost of theft for buyers κ is sufficiently high to satisfy the condition in (28), resulting in $\alpha^b = 0$ and $\alpha^s = 1$ in equilibrium.¹⁶ Then, from (17)-(20) and (28), the inflation rate π and the nominal interest rates on reserves and government bonds R^m and R^b can be expressed as:

$$\pi = \frac{\beta}{\eta} \left[\eta u'(x^c) - \delta u'(x^d) + \delta \right], \quad (30)$$

$$R^m = R^b = \frac{\eta u'(x^c) - \delta u'(x^d) + \delta}{\eta [u'(x^d) - \delta u'(x^d) + \delta]}, \quad (31)$$

and the corresponding real interest rates can be expressed as:

$$r^m = r^b = \frac{1}{\beta [u'(x^d) - \delta u'(x^d) + \delta]}. \quad (32)$$

¹⁶Analyzing an equilibrium where the cost of theft is sufficiently high is analogous to studying a model where the marginal cost of holding currency does not change endogenously.

In this equilibrium, the consumption quantity in DM trades involving deposit claims x^d is inefficiently low due to a binding collateral constraint, leading to low real interest rates.

From (6), (24)-(26), and (30)-(31), the binding collateral constraint (15) can be rewritten as:

$$\left[u'(x^c) + \frac{\delta}{(1-\delta)\eta} \right] \rho(x^c + \beta\mu) + \left[u'(x^d) + \frac{\delta}{1-\delta} \right] (1-\rho)x^d = v. \quad (33)$$

Equation (33) shows that the aggregate demand for collateral (on the left-hand side) must equal its aggregate supply (on the right-hand side) in equilibrium. From the condition in (28), a necessary condition for buyers not to invest in the theft technology can be rewritten as:

$$\kappa \geq \frac{\rho(x^c + \beta\mu)}{\beta}. \quad (34)$$

Finally, the no-arbitrage condition from holding currency across periods can be written as:

$$R^m \geq \max \left\{ \frac{1}{\eta + \beta\gamma}, \frac{\eta}{(\eta + \beta\gamma)[(1-\delta)u'(x^d) + \delta]} \right\} \equiv ELB. \quad (35)$$

If the non-par exchange rate for currency withdrawals η is sufficiently close to one and the real interest rate $r^m = r^b$ is sufficiently low, such that the term $(1-\delta)u'(x^d) + \delta$ is sufficiently high, then the ELB is determined by the first argument. In this case, a higher non-par exchange rate η implies a lower ELB, which aligns with the claims made by [Eisler \(1932\)](#), [Buiter \(2010\)](#), and [Agarwal and Kimball \(2015\)](#). However, if η or $r^m = r^b$ is sufficiently high, then the second argument governs the ELB on the nominal interest rate. In this case, an increase in η does not necessarily lower the ELB. This happens because there are insufficient frictions to prevent sellers from holding currency across periods. Essentially, the marginal cost of holding currency across periods perceived by sellers does not catch up with an increase in its marginal benefit. So, sellers can exploit arbitrage by purchasing currency at price η , holding it across periods, and selling it at the same price without risk of theft. Assuming that the real interest rate remains the same, an increase in η only increases the market price of currency, rendering arbitrage more profitable. Therefore, the ELB might even increase in response to an increase in η .

An interpretation is that for the nominal rate of return on currency to be negative, there must be currency deposits and withdrawals at different exchange rates in equilibrium. If either one of these activities does not occur, a non-par exchange rate may fail to reduce the rate of return on currency and the ELB. In the equilibrium studied here, the lack of currency deposits by sellers breaks the link between the non-par exchange rate and the nominal rate of return on currency. In contrast, if there is deflation in equilibrium, it is possible that private banks choose not to withdraw currency from the central bank, as detailed in [Appendix A.1](#).

Using the above equilibrium conditions, I can solve the model as follows. First, equations (31) and (33) solve for (x^c, x^d) given monetary policy (R^m, η) and fiscal policy v . Then, equation (30) solves for π , equation (32) solves for r^m and r^b , and inequalities (34) and (35) provide necessary

conditions for this equilibrium to exist.

Effects of Monetary Policy Note that the size of the central bank's balance sheet ω is irrelevant to asset prices or consumption quantities. Adjusting the balance sheet size entails the central bank's asset swaps between reserves and government bonds, which in turn affect the composition of assets held by private banks. However, since reserves and government bonds are perfect substitutes at the margin, changes in the balance sheet size have no impact beyond altering the composition of assets in private bank portfolios.

In what follows, I will examine the effects of monetary policy interventions in cases where both the nominal interest rate on reserves R^m and the non-par exchange rate for currency withdrawals η are close to one. This implies that the ELB on the nominal interest rate is determined by the first argument in (35).

Proposition 1 *Suppose that inequalities (34) and (35) hold in equilibrium, and that (R^m, η) is sufficiently close to $(1, 1)$, while the fixed cost of holding currency μ is sufficiently close to zero. Then, an increase in R^m results in a decrease in x^c , an increase in x^d , an increase in real interest rates (r^m, r^b) , and an increase in π , with no effect on the ELB. In contrast, an increase in η leads to an increase in x^c and a decrease in the ELB. Furthermore, there exists a threshold $\hat{v}$ such that for $v \in (0, \hat{v}]$, an increase in η decreases x^d and (r^m, r^b) , while increasing π . Conversely, for $v \in (\hat{v}, \bar{v})$, where $\bar{v}$ is the upper bound for the values of consolidated government debt supporting an equilibrium with a binding collateral constraint, an increase in η increases x^d and (r^m, r^b) , while decreasing π .*

Proof See Appendix $\square$

With the non-par exchange rate for currency withdrawals η held constant, an increase in the nominal interest rate on reserves R^m encourages private banks to hold more reserves and government bonds compared to currency. This effectively increases the real value of collateral held by banks, allowing them to issue a larger quantity of deposit claims to buyers. As a result, the consumption quantity in DM trades involving deposit claims x^d increases. In contrast, the decrease in the real quantity of currency outstanding leads to a decrease in the consumption quantity in DM trades involving currency x^c . This decrease in currency outstanding must also be accompanied by a fall in the real rate of return on currency in equilibrium, implying that the inflation rate π must rise. Additionally, the increase in the real value of collateral acts to relax the collateral constraint faced by banks, leading to a fall in the liquidity premium. Consequently, real interest rates r^m and r^b rise (but by less than the rise in nominal interest rates).

A novel finding is that an increase in the non-par exchange rate for currency withdrawals η itself has real effects. Specifically, with R^m held constant, an increase in η leads to an increase in the consumption quantity in DM trades using currency x^c . This occurs because the increase in η increases the price of currency in the CM, thereby increasing its value in DM trades. The effect of this increase in η on the consumption quantity in DM trades using bank claims x^d depends on the

(For a sufficiently low v)						(For a sufficiently high v)					
	∂x^c	∂x^d	$\partial \pi$	∂r^m	∂ELB		∂x^c	∂x^d	$\partial \pi$	∂r^m	∂ELB
∂R^m	—	+	+	+	·	∂R^m	—	+	+	+	·
$\partial \eta$	+	—	+	—	—	$\partial \eta$	+	+	—	+	—

Table 1: Effects of monetary policy (R^m, η) in an equilibrium with no theft

value of consolidated government debt v . If v is sufficiently low, then x^d decreases. This suggests that a larger quantity of currency outstanding effectively reduces the stock of government bonds and reserves held by private banks, making them to reduce the quantity of deposit claims offered to buyers. In contrast, if v is sufficiently high but not too high, then x^d increases. In this case, the higher price of currency acts to reduce the real quantity of currency c' (as the income effect exceeds the substitution effect), effectively relaxing the collateral constraint in (22). Therefore, both x^c and x^d increase. These findings are summarized in Table 1.

Corollary 1 *If η is sufficiently close to one, an increase in η leads to a decrease in the ELB on the nominal interest rate. However, if η is sufficiently high, an increase in η can increase the ELB.*

As mentioned earlier, when the non-par exchange rate η is sufficiently close to one, the ELB on the nominal interest rate is determined by the first argument in (35). That is, a lack of arbitrage opportunities for private banks from holding currency across periods implies that sellers also lack such opportunities. In this case, the central bank can reduce the ELB below the base lower bound, which is defined as the level of the ELB observed with a typical one-to-one exchange rate between currency and reserves. However, when η is sufficiently high, the second argument in (35) may determine the ELB instead. In this case, the lack of arbitrage opportunities for private banks does not necessarily prevent arbitrage activities for sellers. Sellers can carry currency and sell it at the market price η without facing the threat of theft. As a result, an increase in η may not effectively reduce the rate of return on currency. It is possible that an increase in η leads to a rise in the rate of return on currency perceived by sellers, because the cost of storing currency becomes relatively smaller as the price of currency η rises. This relationship between the non-par exchange rate η and the ELB is depicted by Figure 2.

Corollary 2 *Given that (R^m, η) is sufficiently close to $(1, 1)$ and the fixed cost of holding currency μ is sufficiently close to zero, consider a policy where the central bank increases η and decreases R^m to maintain ηR^m constant. This policy results in an increase in x^c and a decrease in the ELB. Furthermore, if $v \in (0, \hat{v}]$, then x^d decreases and (r^m, r^b) decrease, while the effect on π is ambiguous. If $v \in (\hat{v}, \bar{v})$, then π falls, while the effects on x^d and (r^m, r^b) are ambiguous.*

If the central bank aims to reduce the nominal interest rate R^m below the base lower bound, it can achieve this by altering the ELB. Corollary 2 explores what happens if the central bank increases the non-par exchange rate η and reduces the nominal interest rate on reserves R^m by the

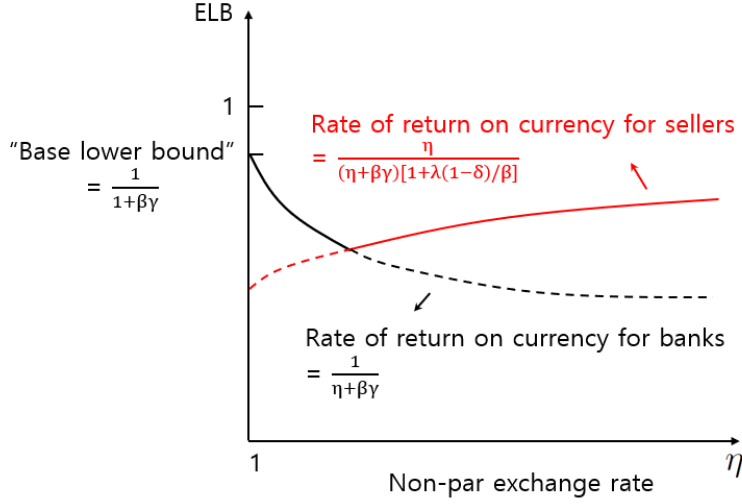


Figure 2: Exchange rate η and the effective lower bound (ELB)

same magnitude, while keeping the relative rate of return on currency $\frac{1}{\eta R^m}$ constant. This policy leads to an increase in x^c and a decrease in the ELB. However, its effects on other variables, including x^d , r^m , r^b , and π , vary depending on the value of consolidated government debt v . Importantly, this policy is not neutral to the economy, as it affects DM consumption quantities and real interest rates. These real effects arise because the expected payoff for sellers from side-trading currency with private banks exceeds the payoff from depositing it with the central bank. Thus, the expected rate of return on currency perceived by sellers does not align with the target rate set by the central bank, leading to a distortion in the allocation of currency and deposit claims in DM trades.

3.6 Equilibrium with Theft

In this section, I analyze an equilibrium where some sellers carry currency into the TM, and some buyers steal currency. Suppose that the cost of acquiring the theft technology κ is sufficiently low to satisfy the condition in (29) in equilibrium. Then, from (17)-(20) and (29), I obtain:

$$\pi = \frac{\beta}{\eta} \left[u'(x^c) - \delta u'(x^d) + \delta \right], \quad (36)$$

$$R^m = R^b = \frac{u'(x^c) - \delta u'(x^d) + \delta}{\eta [u'(x^d) - \delta u'(x^d) + \delta]}, \quad (37)$$

$$r^m = r^b = \frac{1}{\beta [u'(x^d) - \delta u'(x^d) + \delta]}. \quad (38)$$

From (6), (24)-(26), and (36)-(37), the binding collateral constraint (15) can be rewritten as

$$\left[u'(x^c) + \frac{\delta}{1-\delta} \right] \rho(x^c + \beta\mu) + \left[u'(x^d) + \frac{\delta}{1-\delta} \right] (1-\rho)x^d = v. \quad (39)$$

From Lemma 1 and (29), the fraction of buyers who choose to acquire the theft technology α^b and the fraction of sellers who choose to carry currency in the TM α^s can be expressed as:

$$\alpha^b = \frac{\eta - 1}{\eta}, \quad (40)$$

$$\alpha^s = \frac{\beta\kappa}{\rho\eta(x^c + \beta\mu)}, \quad (41)$$

and from the condition in (29), a necessary condition for buyers to invest in the theft technology can be rewritten as:

$$\kappa < \frac{\rho\eta(x^c + \beta\mu)}{\beta}. \quad (42)$$

Finally, from (27) and (40), the no-arbitrage condition from holding currency across periods can be written as:

$$R^m \geq \frac{1}{\eta + \beta\gamma} \equiv ELB. \quad (43)$$

Solving the model is straightforward, similar to an equilibrium with no theft. First, equations (37) and (39) solve for (x^c, x^d) , given monetary policy (R^m, η) and fiscal policy v . Then, equation (36) solves for π , equation (38) solves for r^m and r^b , equations (40) and (41) solve for α^b and α^s , and inequalities (42) and (43) provide necessary conditions for this equilibrium to exist.

Effects of Monetary Policy The size of the central bank's balance sheet ω remains irrelevant to asset prices or consumption quantities, as in an equilibrium without theft. Therefore, I proceed to examine the effects of monetary policy interventions with R^m (the nominal interest rate on reserves) and η (the non-par exchange rate for currency withdrawals).

Proposition 2 *Suppose that inequalities (42) and (43) hold in equilibrium. Then, an increase in either R^m or η results in a decrease in x^c , an increase in x^d , and a rise in real interest rates (r^m, r^b) . In addition, an increase in R^m increases π and α^s , with no effects on the ELB and α^b . Instead, an increase in η decreases π and the ELB, while increasing α^b . However, its effect on α^s is ambiguous.*

Proof See Appendix $\square$

With the non-par exchange rate η held constant, the effects of an increase in the nominal interest rate R^m on consumption quantities (x^c, x^d) , real interest rates (r^m, r^b) , and the inflation rate π are qualitatively identical to those in an equilibrium with no theft. However, in this equilibrium, the fraction of sellers who carry currency into the TM α^s increases. This occurs because, in response to an increase in R^m , sellers receive a smaller quantity of currency from buyers in the DM. Because a smaller quantity of currency carried by sellers discourages buyers from stealing currency, sellers

	∂x^c	∂x^d	$\partial \pi$	∂r^m	∂ELB	$\partial \alpha^b$	$\partial \alpha^s$
∂R^m	−	+	+	+	·	·	+
$\partial \eta$	−	+	−	+	−	+	?

Table 2: Effects of monetary policy (R^m, η) in an equilibrium with theft

can increase the probability of carrying currency into the TM until the fraction of buyers investing in the theft technology α^b reaches the previous level.

A key result is that an increase in the non-par exchange rate η (which effectively decreases the nominal/real rate of return on currency) leads to a decrease in the inflation rate π (which increases the real rate of return on currency). So, the fall in the real rate of return on currency due to an increase in η is mitigated by a decrease in π in equilibrium. Also, an increase in η decreases the ELB on nominal interest rates. These results are consistent with the ideas presented by [Eisler \(1932\)](#), [Buiter \(2010\)](#), and [Agarwal and Kimball \(2015\)](#), in that introducing a non-par exchange rate between currency and reserves can reduce the nominal rate of return on currency and the ELB.

A higher η , or a higher price of currency in the CM, not only induces sellers to carry more currency into the TM but also encourages more buyers to invest in the theft technology. So, despite the increase in the marginal benefit of side-trading currency, the increased risk of theft raises its marginal cost, making sellers indifferent between side-trading currency with private banks and safely depositing it with the central bank.¹⁷ As a result, the central bank can effectively reduce the rate of return on currency and the ELB due to endogenous theft. However, reducing the ELB comes at a cost, because a larger fraction of buyers investing in the theft technology implies a greater welfare loss.

Similar to what happens in an equilibrium with no theft, an increase in the non-par exchange rate η itself has real effects. Specifically, it decreases x^c and increases x^d , r^m and r^b . These effects arise because, with R^m held constant, an increase in η leads to a fall in the rate of return on currency relative to government bonds and reserves $\frac{1}{\eta R^m}$. Note that the effects of an increase in the non-par exchange rate η on consumption quantities and real interest rates are qualitatively the same as those of an increase in R^m . This is because both policies serve to reduce the rate of return on currency relative to interest-bearing assets (government bonds and reserves). These results are summarized in [Table 2](#).

Corollary 3 *Consider a policy where the central bank increases η and decreases R^m to keep ηR^m constant. This policy decreases π one-for-one, decreases α^s and the ELB, and increases α^b . However, consumption quantities (x^c, x^d) and real interest rates (r^m, r^b) remain unchanged.*

If the central bank aims to reduce the nominal interest rate R^m below the base lower bound, it needs to reduce the ELB on the nominal interest rate. Suppose that the central bank increases

¹⁷The effect of an increase in η on sellers' behavior is ambiguous. While a greater marginal benefit tends to increase the fraction of sellers carrying currency into the TM α^s , a greater marginal cost tends to reduce it. Thus, the effect on α^s remains ambiguous.

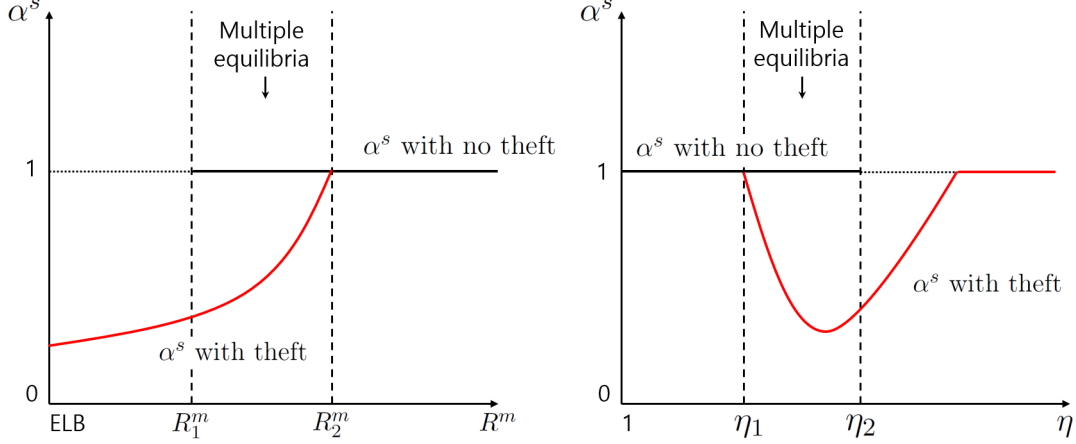


Figure 3: Equilibrium α^s with monetary policy (R^m, η)

the non-par exchange rate η and reduces the nominal interest rate on reserves R^m by the same magnitude to maintain the relative rate of return on currency $\frac{1}{\eta R^m}$ constant. Corollary 3 states that this policy decreases the inflation rate π one-for-one with an increase in η , implying no effect on the real rate of return on currency. Also, real interest rates (r^m, r^b) remain unaffected because the decrease in π offsets the decrease in R^m one-for-one (a pure Fisher effect). Consequently, there are no effects on consumption quantities (x^c, x^d) . However, the fraction of buyers acquiring the theft technology α^b increases, which has welfare implications.

With the necessary conditions (34) and (42), Propositions 1 and 2 help us understand what type of equilibrium (or equilibria) may arise under different monetary policies (R^m, η) . When η remains constant, an increase in R^m decreases x^c , implying that an equilibrium with no theft is more likely to arise for higher values of R^m . Further, if η is sufficiently high or the cost of theft κ is sufficiently low, theft occurs for sufficiently low values of R^m . In such a case, there may be a range of nominal interest rates R^m , as depicted in the left panel of Figure 3, that support equilibria both with and without theft. This means that, for some parameter values, multiple equilibria can emerge. In the figure, the upper bound R_2^m for nominal interest rates supporting *theft* equilibrium is higher than the lower bound R_1^m for rates sustaining *no-theft* equilibrium. Thus, multiple equilibria exist for $R^m \in [R_1^m, R_2^m]$. However, for some parameter values, the upper bound R_2^m can be lower than the lower bound R_1^m , implying the absence of equilibrium for $R^m \in (R_2^m, R_1^m)$.

Propositions 1 and 2 also state that, with R^m held constant, an increase in η decreases x^c in theft equilibria, while increasing x^c in no-theft equilibria. This implies that a no-theft equilibrium is more likely to arise for lower values of η and higher values of κ (the cost of theft). Specifically, if either R^m or κ is sufficiently high, theft does not occur for sufficiently low values of η . In such cases, multiple equilibria may exist for sufficiently high but not too high values of η , as illustrated in the right panel of Figure 3. Note that the effect of an increase in η on α^s (the fraction of sellers carrying currency into the TM) is ambiguous in a theft equilibrium. The right panel of Figure

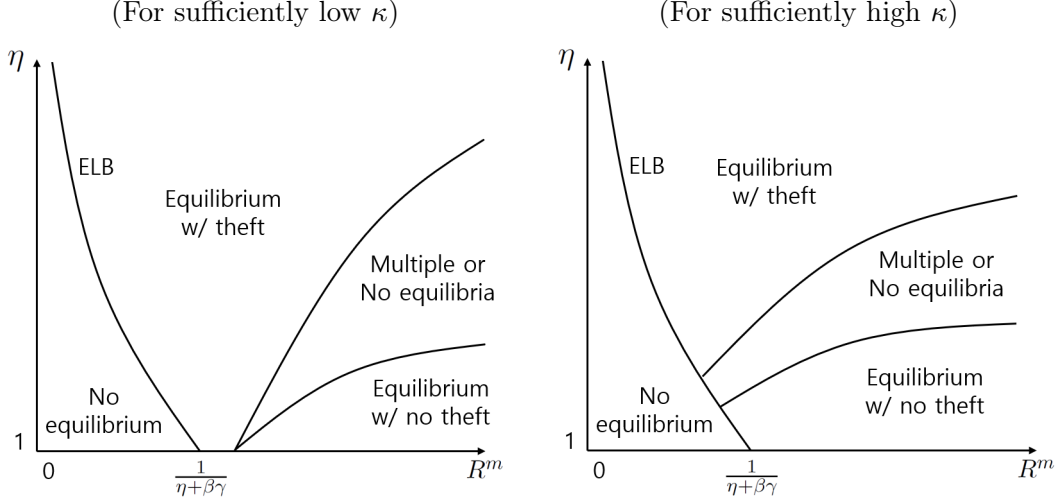


Figure 4: Equilibria with monetary policy (R^m, η)

3 presents a special case where an increase in η decreases α^s for low values of η and increases it for high values of η . Moreover, if η is very high, it is possible that all currency-holding sellers carry their currency into the TM ($\alpha^s = 1$), even though some buyers steal currency in equilibrium ($0 < \alpha^b < 1$). In this section, the analysis has focused on a theft equilibrium where sellers are indifferent between depositing currency and not depositing, i.e., $0 < \alpha^b < 1$ and $0 < \alpha^s < 1$. Lastly, Figure 4 illustrates how the nominal interest rate on reserves R^m and the non-par exchange rate η determine the existence of particular equilibria.

4 Optimal Monetary Policy

I define welfare as

$$\mathcal{W} = \underbrace{0}_{\text{CM surpluses}} + \underbrace{\rho [u(x^c) - x^c] + (1 - \rho) [u(x^d) - x^d]}_{\text{DM surpluses}} - \underbrace{\alpha^b \kappa}_{\text{total cost in TM}}, \quad (44)$$

which is the sum of surpluses from trade in the CM and the DM, net of the total cost incurred in the TM. Welfare defined here is also equivalent to the sum of period utilities across agents in equilibrium. I will discuss the optimal monetary policy using this welfare measure.

Proposition 3 *If the cost of theft κ is sufficiently low and the fixed cost of holding currency μ is sufficiently close to zero, then the optimal monetary policy consists of $\eta = 1$ and $R^m \leq 1$. Moreover, in cases where the optimal nominal interest rate is constrained by the base lower bound, the optimal monetary policy consists of $\eta = 1$ and $R^m = \text{base lower bound}$.*

Proof See Appendix $\square$

To understand the intuition behind these results, consider cases where there is no fixed cost associated with holding currency at the end of the DM ($\mu = 0$) as a benchmark. In Appendix A.3, I formalize the proof of Proposition 3 in two steps. Firstly, taking α^b (the fraction of buyers acquiring the theft technology) as given, an optimal monetary policy can be characterized by a *modified* Friedman rule, meaning that the nominal interest rate on reserves relative to currency is zero (or equivalently, $\eta R^m = 1$). A modified Friedman rule achieves a social optimum, given α^b , by allowing buyers to perfectly smooth their DM consumption across different states of the world. Secondly, among the policy alternatives compliant with this rule, I find that the optimal monetary policy consists of $\eta = 1$ and $R^m = 1$. This policy combination minimizes the resource costs incurred due to the socially useless theft technology, while maximizing the surpluses from trade.¹⁸

Finally, when holding currency incurs a fixed cost at the end of the DM ($\mu > 0$), a negative nominal interest rate ($R^m < 1$) is optimal, given a one-to-one exchange rate $\eta = 1$. This is primarily because buyers need to bring more currency into the DM to compensate for the currency-holding costs incurred by sellers. The presence of this fixed cost distorts the allocation of currency and deposit claims, leading to inefficiencies in DM trades. Therefore, a negative nominal interest rate helps alleviate these inefficiencies by indirectly reducing the cost of holding currency, thus improving welfare.

It is worth noting that even if the optimal interest rate is constrained by the base lower bound, introducing a non-par exchange rate does not help enhance welfare. According to Corollary 3, such a policy only increases costly theft without increasing the surpluses from trade in both the DM and the CM. Therefore, it remains optimal to set the nominal interest rate at the base lower bound.

Proposition 4 *If the cost of theft κ is sufficiently high, then the optimal monetary policy is given by $\eta = \bar{\eta}$, where $\bar{\eta}$ is the solution to (31) and (33) for $x^c = \beta(\frac{\kappa}{\rho} - \mu)$. Moreover, if the fixed cost of holding currency μ is sufficiently close to zero, then the optimal nominal interest rate on reserves R^m satisfies $R^m > 1/\bar{\eta}$.*

Proof See Appendix $\square$

Interestingly, when the cost of theft κ is sufficiently high to prevent theft in equilibrium, the optimal nominal interest rate can be higher than the level that satisfies a modified Friedman rule. In Appendix A.3, I show that deviating from the modified Friedman rule by increasing either the nominal interest rate R^m or the non-par exchange rate η can improve social welfare, i.e., the optimal policy can be characterized by $\eta R^m > 1$. An increase in R^m enhances welfare by enabling buyers to smooth their consumption across different DM meetings. However, increasing η from the modified Friedman rule enhances welfare for a different reason. In a no-theft equilibrium, an increase in η

¹⁸Optimality is achieved when the exchange rate between currency and reserves is one-to-one, as in a traditional central banking system, and the nominal interest rate on reserves is zero (a classic Friedman rule). Interestingly, even when negative nominal interest rates are available, a zero nominal interest rate remains optimal. This zero nominal interest rate can be considered as the "reversal interest rate," the interest rate at which lowering the interest rate becomes contractionary (Abadi, Brunnermeier, and Koby, 2023).

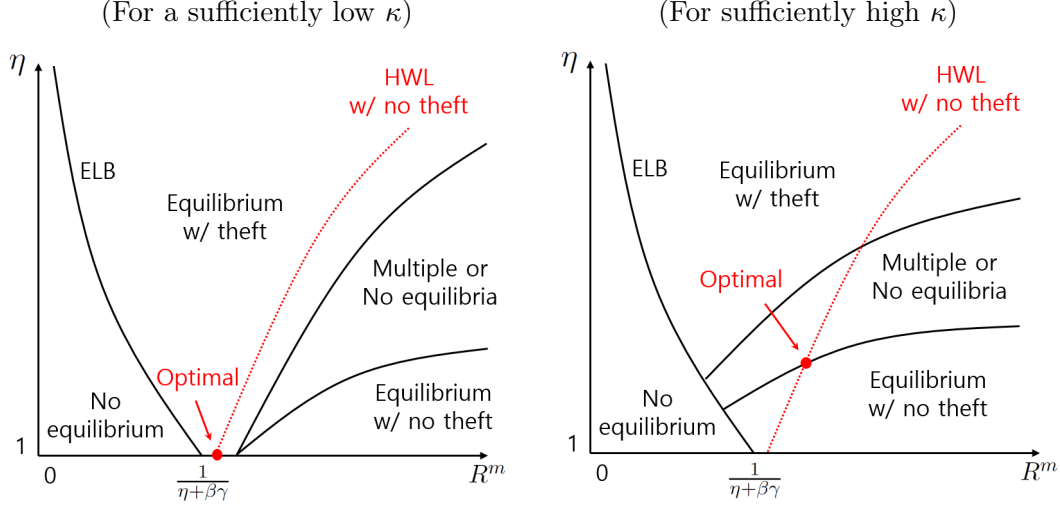


Figure 5: Optimal monetary policy

does not effectively reduce the perceived nominal rate of return on currency for sellers, while raising the price of currency in the CM. This increases the value of currency in the DM, leading to an increase in consumption in DM meetings involving currency x_c . Although this may be accompanied by a decrease in consumption in DM meetings involving deposit claims x_d , the potential decrease in the buyers' utility from the change in x_d is smaller than the increase in their utility from the increase in x_c . Therefore, an increase in η can improve welfare even if it does not smooth out consumption across different DM meetings. Furthermore, as welfare increases with the non-par exchange rate η , it is optimal to set η at the highest possible level that does not induce theft in equilibrium.

These results can be visually represented by Figure 5. Each panel in the figure illustrates the Highest Welfare Locus (HWL), which shows the combinations of nominal interest rates R^m and non-par exchange rates η that yield the highest welfare, assuming that no theft occurs in equilibrium. In the left panel, where the cost of theft κ is sufficiently low, welfare is maximized at $\eta = 1$ and $R^m < 1$. Conversely, in the right panel, where κ is sufficiently high, welfare is maximized at the highest possible level of η along with the corresponding R^m on the HWL that supports a no-theft equilibrium as the unique equilibrium.

5 Disintermediation

A concern surrounding the implementation of negative nominal interest rates is the possibility of disintermediation, where consumers withdraw all currency from their deposit accounts.¹⁹ However, in the baseline model analyzed in previous sections, disintermediation does not occur because currency and bank deposits are not substitutes. Specifically, I assumed that only currency is accepted in some

¹⁹Disintermediation is indeed a practical concern. As a negative deposit rate might lead to massive cash withdrawals, private banks are reluctant to reduce their deposit rates below zero. See Eggertsson, Juelsrud, Summers, and Wold (2022) for empirical evidence on the breakdown of the monetary policy transmission when the policy rate is negative.

transactions, while only bank deposit claims are accepted in others. To explore the implications of introducing a non-par exchange rate between currency and reserves for potential disintermediation, I modify the baseline model by allowing currency to be acceptable in all DM transactions. Although this assumption is extreme, it enables us to examine how a non-par exchange rate between currency and reserves can help prevent disintermediation.

In the modified model, buyers may indeed find it optimal to opt out of banking arrangements since currency is accepted everywhere. In the previous sections, I assumed that a fraction $1 - \rho$ of sellers accept only bank deposit claims as a means of payment in DM transactions. Now, I will assume that these sellers accept both currency and deposit claims, while a fraction ρ of sellers accept only currency as in the baseline model. Other than this modification, the model remains the same as described in Section 2.

Let θ denote the fraction of buyers who choose to deposit with private banks in the CM. Each private bank's contracting problem and the first-order conditions for the bank's problem are identical to those in the baseline model. Then, a fraction $1 - \theta$ of buyers choose to opt out of banking arrangements and use only currency in the DM. Each of these buyers chooses the quantity of currency that solves:

$$\max_{c^o \geq 0} \left\{ -\eta c^o + u \left(\frac{\beta c^o [1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta]}{\pi} - \beta\mu \right) \right\}, \quad (45)$$

where c^o is the real quantity of currency each buyer obtains in the CM. Although private banks do not write deposit contracts with these buyers, they withdraw currency from the central bank to sell it to these buyers whenever necessary. Then, no arbitrage implies that the price of currency is η in equilibrium. The first-order condition for each of these buyer's problem is given by:

$$-\eta + \frac{\beta [1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta]}{\pi} u' \left(\frac{\beta c^o [1 - \alpha^s + \alpha^s(1 - \alpha^b)\eta]}{\pi} - \beta\mu \right) = 0. \quad (46)$$

and let x^o denote the consumption quantity in the DM for the buyer. Asset market clearing conditions are given by:

$$\theta c + (1 - \theta)c^o = \bar{c}; \quad \theta m = \bar{m}; \quad \theta b = \bar{b}. \quad (47)$$

Let U^b denote the expected utility for buyers who obtain banking contracts, and let U^o denote the expected utility for buyers who opt out of banking contracts. Then, using (14), (17)-(19), and (46), U^b and U^o can be written as:

$$U^b = \rho [u(x^c) - (x^c + \beta\mu)u'(x^c)] + (1 - \rho) [u(x^d) - x^d u'(x^d)], \quad (48)$$

$$U^o = u(x^o) - (x^o + \beta\mu)u'(x^o). \quad (49)$$

In equilibrium, the fraction θ solves the following problem:

$$\max_{0 \leq \theta \leq 1} [\theta U^b + (1 - \theta) U^o], \quad (50)$$

which implies that θ must be consistent with each buyer's utility maximization problem.

Depending on whom each seller met in the previous DM, the quantity of currency holdings in the TM vary across sellers. Buyers who hold deposit contracts withdraw c' units of currency, in real terms, to purchase goods in the DM. Thus, sellers who meet these buyers in the DM will hold $\frac{c'}{\pi}$ units of real currency in the following TM. However, buyers who do not hold deposit contracts have c^o units of currency. So, sellers who meet these buyers in the DM will hold $\frac{c^o}{\pi}$ units of real currency in the following TM. For simplicity, I assume that each seller decides whether to carry currency into the TM before learning the type of the buyer they meet in the DM.

The fraction of buyers acquiring the theft technology α^b and the fraction of sellers carrying their currency into the TM α^s must be incentive-compatible in equilibrium. Then,

$$\text{if } \kappa > \frac{[\theta \rho c' + (1 - \theta) c^o] \alpha^s \eta}{\pi}, \quad \text{then } \alpha^b = 0, \quad (51)$$

$$\text{if } \kappa = \frac{[\theta \rho c' + (1 - \theta) c^o] \alpha^s \eta}{\pi}, \quad \text{then } 0 \leq \alpha^b \leq 1, \quad (52)$$

$$\text{if } \kappa < \frac{[\theta \rho c' + (1 - \theta) c^o] \alpha^s \eta}{\pi}, \quad \text{then } \alpha^b = 1. \quad (53)$$

With probability $\theta \rho \alpha^s$, each buyer is matched with a seller holding $\frac{c'}{\pi}$ units of real currency, and with probability $(1 - \theta) \alpha^s$ each buyer is matched with a seller holding $\frac{c^o}{\pi}$ units of real currency. Conditions (51)-(53) state that theft does not occur if it is too costly, it sometimes takes place if the buyer is indifferent between stealing and not stealing, and it always takes place if it is profitable. Also, each seller's decision on whether to carry currency into the TM must be incentive-compatible. That is,

$$\text{if } (1 - \alpha^b) \eta < 1, \quad \text{then } \alpha^s = 0, \quad (54)$$

$$\text{if } (1 - \alpha^b) \eta = 1, \quad \text{then } 0 \leq \alpha^s \leq 1, \quad (55)$$

$$\text{if } (1 - \alpha^b) \eta > 1, \quad \text{then } \alpha^s = 1. \quad (56)$$

In this section, I will confine attention to cases where the cost of theft κ is sufficiently small to sustain a theft equilibrium.²⁰ Also, I will consider cases where collateralizable assets are scarce, as in the previous sections. Specifically, I will assume that

$$v < x^* + \beta \mu, \quad (57)$$

where x^* is the efficient level of consumption in DM transactions that satisfies $u'(x^*) = 1$. This

²⁰For cases with a sufficiently high cost of theft, equilibrium conditions can be presented, but analyzing the effects of monetary policy in such cases may not be straightforward.

assumption implies that the consumption quantities in DM transactions for buyers choosing bank contracts are inefficiently low due to the shortage of collateral. However, it also implies that the central bank cannot support the efficient level of consumption for those opting out of banking contracts. This is because the quantity of currency outstanding is constrained by the size of the central bank's balance sheet, which can only be increased by the central bank's purchase of scarce government debt. Given this assumption, the following proposition characterizes the effective lower bound (ELB) on nominal interest rates R^m .

Proposition 5 *Suppose that both the cost of theft κ and the fixed cost of holding currency μ are sufficiently low, and that the value of consolidated government debt outstanding v satisfies (57). Then, an equilibrium exists if and only if*

$$R^m \geq \frac{u'(x^o)}{\eta[(1 - \delta)u'(x^d) + \delta]}, \quad (58)$$

where (x^o, x^d) , together with x^c , consist of the solution to $(x^o + \beta\mu)u'(x^o) = v$, $u'(x^o) = u'(x^c) - \delta u'(x^d) + \delta$, and $U^b = U^o$ from (48)-(49). Furthermore,

$$\frac{u'(x^o)}{(1 - \delta)u'(x^d) + \delta} \geq 1.$$

Proof See Appendix $\square$

According to Proposition 5, the nominal interest rate R^m cannot fall below the threshold presented in (58) in equilibrium. If R^m falls below this threshold, all buyers would prefer to use currency in every DM meeting ($\theta = 0$). However, the central bank would not be able to meet the public demand for currency due to the shortage of government bonds it can purchase from the public. Since nominal interest rates lower than the threshold cannot be sustained in equilibrium, the threshold acts as the ELB on nominal interest rates in this economy. Furthermore, introducing a non-par exchange rate for currency withdrawals $\eta > 1$ reduces this threshold, thereby lowering the ELB. This result is consistent with the ideas proposed by [Eisler \(1932\)](#), [Buiter \(2010\)](#), and [Agarwal and Kimball \(2015\)](#).

In this model, the proportional cost of storing currency γ becomes irrelevant because nominal interest rates that encourage buyers to participate in banking arrangements are sufficiently high to prevent arbitrage opportunities from storing currency across periods.²¹ Interestingly, the ELB on nominal interest rates can even be positive. This occurs due to collateral scarcity, which creates

²¹This finding is related to [Eggertsson, Juelsrud, Summers, and Wold \(2022\)](#), in that the lower bound on deposit rates (disintermediation-free interest rates) tends to be higher than the lower bound on short-term policy rates (arbitrage-free interest rates). While my model abstracts from addressing the conflict between these different lower bounds, it suggests that introducing a non-par exchange rate policy could reduce both lower bounds by decreasing the rate of return on holding currency for both banks and depositors.

inefficiency in the banking system. When the nominal interest rate on reserves is zero ($R^m = 1$), each contracting buyer would be able to consume the same quantity of goods across different DM meetings, i.e., $x^c = x^d$. However, if the fixed cost of holding currency μ is sufficiently low, a non-contracting buyer's consumption in the DM would be larger than that for a contracting buyer, i.e., $x^o > x^c = x^d$, leading to complete disintermediation ($\theta = 0$). Since such a scenario is not sustainable in equilibrium, the nominal interest rate must be higher than zero to support the banking system.²²

From now on, I will focus on cases where $R^m > \frac{u'(x^o)}{\eta[u'(x^d) - \delta u'(x^d) + \delta]}$ and assume that $\mu = 0$ (no fixed cost of holding currency at the beginning of the TM) for analytical convenience.²³ Then, from (6), (17)-(22), and (46)-(47), I derive:

$$\eta R^m = \frac{u'(x^c) - \delta u'(x^d) + \delta}{u'(x^d) - \delta u'(x^d) + \delta}, \quad (59)$$

$$(1 - \rho)\theta x^d \left[u'(x^d) + \frac{\delta}{1 - \delta} \right] + \rho\theta(x^c) \left[u'(x^c) + \frac{\delta}{1 - \delta} \right] + (1 - \theta)(x^o)u'(x^o) = v, \quad (60)$$

$$\pi = \frac{\beta [u'(x^c) - \delta u'(x^d) + \delta]}{\eta}. \quad (61)$$

Equations (59)-(61) come from the first-order conditions for a private bank's problem in equilibrium, with (60) representing the collateral constraint. Notably, this collateral constraint differs somewhat from that in the baseline model (represented by 39), as it accounts for the central bank's purchases of government bonds when issuing currency for non-contracting buyers. From (17), (18), (46), and (48)-(50),

$$u'(x^o) = u'(x^c) - \delta u'(x^d) + \delta, \quad (62)$$

$$\rho [u(x^c) - x^c u'(x^c)] + (1 - \rho) [u(x^d) - x^d u'(x^d)] \geq u(x^o) - x^o u'(x^o). \quad (63)$$

Equation (62) is a necessary condition that ensures positive consumption quantities in the DM for both contracting and non-contracting buyers. This equation also tells us that the choices of buyers must be incentive-compatible in equilibrium. Note that collateral scarcity, which creates inefficiencies in deposit contracts, not only constrains x^d but also x^c . Thus, buyers opt for deposit contracts only when $x^c < x^d$, as they would otherwise prefer to opt out. Additionally, if $x^o < x^c$, no buyers would opt out of deposit contracts; conversely, if $x^o > x^d$, no buyers would opt in. Therefore, the consumption quantities in the DM must satisfy $x^c < x^o < x^d$ in equilibrium, as indicated by (62).²⁴ Equation (63) further shows that, in equilibrium, buyers must weakly prefer opting for a

²²One can imagine that a higher fixed storage cost μ would increase inefficiency in DM meetings involving currency. This would discourage buyers from opting out of deposit contracts, thereby lowering the ELB. While analyzing the qualitative effect of an increase in μ appears to be complicated, the intuition suggests that a sufficiently high μ would lead to a negative ELB on nominal interest rates.

²³This assumption simplifies the analysis while still providing insights that can be extended to cases with sufficiently low $\mu > 0$.

²⁴However, using only currency in the DM is also inefficient due to the fixed cost of holding currency μ . With a sufficiently high μ , buyers could choose to use deposit contracts even when $x^d \leq x^c$.

deposit contract over opting out. Finally, from (52) and (55),

$$\alpha^b = \frac{\eta - 1}{\eta}, \quad (64)$$

$$\alpha^s = \frac{\beta \kappa}{\eta [\theta \rho x^c + (1 - \theta) x^o]}, \quad (65)$$

$$\kappa < \frac{\eta [\theta \rho x^c + (1 - \theta) x^o]}{\beta}. \quad (66)$$

Equations (64) and (65) determine the fraction of buyers acquiring the theft technology α^b and the fraction of sellers carrying their currency into the TM α^s . Inequality (66) is a necessary condition for this equilibrium to exist.

The model can be solved differently depending on whether (63) holds with equality. If (63) holds with equality, equations (59), (62), and (63) solve for x^c , x^d , and x^o given monetary policy (R^m, η) and fiscal policy v . Then, equation (60) solves for θ , equation (61) solves for π , and equations (64) and (65) solve for α^b and α^s , respectively. If (63) holds with strict inequality, equations (59) and (60) with $\theta = 1$ solve for x^c and x^d . Then, equation (61) solves for π and equations (64) and (65) solve for α^b and α^s , respectively, with $\theta = 1$. Equation (62) solves for x^o , the would-be quantity of consumption in DM transactions if a buyer were to opt out of banking arrangements off equilibrium.

The following proposition shows how the fraction θ is determined in equilibrium and the effects of monetary policy (R^m, η) depending on the value of θ .

Proposition 6 *Suppose that the fixed cost of holding currency μ is zero. If ηR^m is sufficiently high but not too high, then $0 \leq \theta < 1$ in equilibrium and (63) holds with equality. In this case, an increase in ηR^m (an increase in R^m or η or both) leads to decreases in x^c and x^o , and increases in x^d and θ , along with an increase in real interest rates (r^m, r^b) . Furthermore, an increase in R^m increases π and α^s but does not affect α^b . An increase in η decreases π and increases α^b , but its effect on α^s is ambiguous. If ηR^m is very high, then $\theta = 1$ in equilibrium and (63) holds with strict inequality.*

Proof See Appendix $\square$

With the non-par exchange rate for currency withdrawals η held constant, an increase in the nominal interest rate R^m decreases the rate of return on currency relative to reserves and government bonds, creating a substitution of deposit claims for currency in DM transactions. If $0 \leq \theta < 1$ in equilibrium, the consumption quantity in DM transactions using deposit claims x^d increases along with a rise in the fraction of buyers opting for deposit contracts θ , while the consumption quantities in DM transactions using currency x^c and x^o decrease. Thus, when the exchange rate between currency and reserves is held constant, lowering the nominal interest rate R^m may contribute to disintermediation by decreasing the fraction of buyers opting for banking arrangements θ .

However, the central bank can offset this effect by introducing a non-par exchange rate for currency withdrawals η , which helps maintain the relative rate of return on currency $\frac{1}{\eta R^m}$ constant. In other words, a non-par exchange rate policy allows the central bank to implement a negative nominal interest rate without causing a disruption in the banking system, aligning with the conventional view. With a policy holding ηR^m constant, fraction θ as well as x^c , x^d , and x^o would remain unchanged. However, it is worth noting that the fraction of buyers investing in the costly theft technology α^b would increase, leading to welfare losses.

The welfare measure for this economy can be defined as:

$$\mathcal{W} = \rho\theta [u(x^c) - x^c] + (1 - \rho)\theta[u(x^d) - x^d] + (1 - \theta)[u(x^o) - x^o] - \alpha^b\kappa. \quad (67)$$

which is the sum of surpluses from trade in the CM and the DM, net of the total cost of theft incurred in the TM. Then, the following proposition characterizes the optimal monetary policy.

Proposition 7 *Suppose that the fixed cost of holding currency μ is zero. Then, the optimal monetary policy consists of $\eta = 1$ and $R^m = \frac{u'(x^o)}{u'(x^d) - \delta u'(x^d) + \delta} > 1$, where (x^o, x^d) , together with x^c , consist of the solution to $x^o u'(x^o) = v$, (62), and (63) with equality.*

Proof See Appendix $\square$

The central bank can optimize the surplus from trade by choosing an appropriate policy combination (η, R^m) . While there exists a set of such combinations that achieve the maximized level of trade surplus, maintaining the one-to-one exchange rate between currency and reserves ($\eta = 1$), as seen in a traditional central banking system, is always optimal. This is because the traditional one-to-one exchange rate enables the central bank to eliminate costly theft without reducing welfare. Furthermore, optimality is achieved when the central bank sets the nominal interest rate on reserves R^m at the base lower bound (the level of the ELB when $\eta = 1$). Given that the base lower bound on the nominal interest rate R^m is above zero, the consumption quantity in DM transactions using deposit claims exceeds those involving currency for any given R^m . Consequently, reducing the nominal interest rate until it reaches the lower bound maximizes welfare by allowing buyers to smooth their consumption across different DM transactions.

When considering real-world payment systems, both the baseline model and the modified model offer valuable insights, yet each captures different aspects. The baseline model effectively reflects the distinction between cash transactions and electronic transactions, where some transactions can only be made with currency due to various reasons such as accessibility or privacy concerns, while others solely involve deposit claims, as seen in online transactions. On the other hand, the modified model addresses the substitutability between currency and deposit claims but does not consider electronic transactions where only deposit claims are accepted. Therefore, neither model fully captures the complexity and diversity of real-world payment systems, but both provide valuable insights into the implications of implementing a non-par exchange rate policy for the ELB and welfare.

6 Conclusion

In the existing literature, a non-par exchange rate between currency and reserves has been proposed as a potential policy tool for reducing the effective lower bound (ELB) on nominal interest rates. To examine the implications of introducing this non-par exchange rate, I have developed a model with multiple means of payment—currency and bank deposits—incorporating frictions associated with the storage and security costs of currency. A key finding is that a non-par exchange rate can indeed reduce the ELB on nominal interest rates, as these frictions incentivize economic agents to maintain their deposit accounts rather than avoiding the banking system.

However, introducing a non-par exchange rate may entail welfare costs by intensifying the frictions that determine the ELB.²⁵ Specifically, this policy can raise the market value of currency, promoting socially undesirable behaviors like currency side-trading and theft, ultimately leading to welfare losses. Importantly, even if the optimal interest rate is constrained by the ELB, implementing a non-par exchange rate may not help increase welfare, as the effects of reducing the ELB can offset the effects of lowering the nominal interest rate. Consequently, the optimal monetary policy involves setting the nominal interest rate at the current ELB level while maintaining the one-to-one exchange rate between currency and reserves to avoid potential welfare losses. Using a modified version of the model, I have also demonstrated how a non-par exchange rate can enable the central bank to implement negative interest rates, albeit at a cost, without inducing disintermediation.

Replacing physical currency with central bank digital currency (CBDC) can also help reduce the ELB. This is because the central bank can directly introduce a negative nominal interest rate on CBDC, a possibility not feasible with physical currency. However, while the central bank can easily reduce the ELB, the economic effects of implementing a negative interest rate on reserves along with a negative interest rate on CBDC remain uncertain. The findings of this paper suggest that such a policy combination may not improve welfare, as the nominal interest rate on reserves relative to CBDC, which determines the monetary policy stance, would remain unchanged.²⁶

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²⁵In Appendix A.2, I calibrate the model to the U.S. economy to quantify the magnitude of the welfare cost incurred by a non-par exchange rate. Compared to the welfare cost of inflation previously identified in the literature, I find that the welfare cost of reducing the ELB is not negligible.

²⁶This conjecture aligns with the findings of Williamson (2022), which demonstrate that in an economy with CBDC, government bonds, and private capital, the relevant interest rate is the interest rate on government bonds relative to CBDC rather than the interest rate on government bonds relative to zero-interest currency.

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A Appendix

A.1 Equilibrium with Deflation and No Theft

In this section, I focus on stationary equilibria characterized by deflation and a sufficiently high cost of theft κ . With deflation, the real value of currency increases over time, allowing private banks to acquire a sufficient amount of currency in the CM. As private banks do not need to withdraw currency from the central bank, they are no longer indifferent between acquiring currency from sellers and withdrawing it from the central bank. Instead, sellers must be indifferent between depositing their currency with the central bank and engaging in side-trading with private banks. Thus, in equilibrium, one unit of real currency is exchanged for one unit of goods in the CM. From (7) and (28), a necessary condition for this equilibrium to exist is given by:

$$\kappa \geq \frac{\rho(x^c + \beta\mu)}{\beta}.$$

As the equilibrium price of real currency is one, instead of η , the values of η in equations (6)-(12), (17)-(22) must be replaced by one. Then, from (19) and (21), the no-arbitrage condition for private

banks from holding currency across periods can be expressed as:

$$R^m \geq \frac{1}{1 + \beta\gamma}.$$

This implies that the non-par exchange rate η is irrelevant to the effective lower bound (ELB). In this case, the non-par exchange rate policy is ineffective because, if it is introduced, private banks would not withdraw currency from the central bank any longer. Also, it is obvious that the non-par exchange rate does not affect equilibrium prices and allocations.

The inflation rate in this equilibrium can be written as

$$\pi = \beta \left[u'(x^c) - \delta u'(x^d) + \delta \right].$$

To observe deflation in equilibrium, both β and $u'(x^c)$ must be sufficiently low, while δ and $u'(x^d)$ must be sufficiently high. It turns out that v must be sufficiently high and R^m is sufficiently low to support deflation in equilibrium. Note that, in the main body of the paper, I focus on cases with sufficiently low v and γ to ensure that there is inflation in equilibrium for any R^m higher than the ELB.

A.2 Quantitative Analysis

Theoretically, introducing a non-par exchange rate between currency and reserves can lead to welfare losses by incentivizing costly theft and distorting the equilibrium allocation. To quantify these welfare losses, I calibrate the baseline model to the U.S. economy and conduct a counterfactual analysis to evaluate the effects of introducing a non-par exchange rate between currency and reserves.

A.2.1 Calibration

I consider an annual model where the utility function in the DM takes the form $u(x) = \frac{x^{1-\sigma}}{1-\sigma}$. When calibrating the baseline model to data, I exclude the cost of theft κ because a one-to-one exchange rate between currency and reserves implies no theft occurring in the model.²⁷ Then, there are eight parameters to calibrate: σ (the curvature of DM consumption), β (the discount factor), ρ (the fraction of currency transactions in the DM), μ (the fixed cost of storing currency), γ (the proportional cost of storing currency), δ (the fraction of assets private banks can abscond with), R^m (the nominal interest rate on reserves), and v (the value of the consolidated government's liabilities held by the public).

Table 3 summarizes the calibration results along with the target moments, which are mostly constructed from U.S. data for 2013-2015. This period is chosen because, for the purpose of this exercise, it is suitable to consider a timeframe when the policy rate was close to zero. Also, key

²⁷To quantify the welfare cost arising from an increase in theft, the cost of theft parameter κ needs to be calibrated outside the model. However, to the best of my knowledge, there is no available data that allows for the direct measurement of this parameter.

Parameters	Values	Calibration targets	Sources
β	0.96	Standard in literature	
R^m	1.0025	Avg. interest rate on reserves: 0.25%	FRED
γ	0.00	Lowest target range for fed funds rate: 0-0.25%	FRED
σ	0.17	Money demand elasticity (1959-2007): -4.19	FRED
ρ	0.17	Currency to M1 ratio: 17.22%	FRED; Lucas and Nicolini (2015)
v	1.13	Avg. locally-held public debt to GDP: 66.73%	FRED
δ	0.45	Avg. inflation rate: 1.06%	FRED
μ	0.01	Fixed storage cost: 2% of currency payments	Author's assumption

Table 3: Calibration results

variables such as the nominal interest rate on reserves and domestically-held public debt to GDP were stable during this period.

There are three parameters calibrated externally. The discount factor β is given by $\beta = 0.96$. From Federal Reserve Economic Data (FRED), the nominal interest rate on reserves was 0.25 percent over the period 2013-2015 ($R^m = 1.0025$). Finally, the lowest target range for the federal funds rate has been between 0 and 0.25 percent since 1954. Although this does not imply that the proportional cost of storing currency is zero, the proportional cost γ is assumed to be zero for convenience.²⁸

Calibrating σ (the curvature of DM consumption) involves matching the elasticity of money demand in the model with the empirical money demand elasticity obtained from the data. Estimating the money demand elasticity requires a longer time series of data, so I select the time period from 1959 to 2007.²⁹ Using data on currency in circulation and nominal GDP (from FRED), I calculate the currency-to-GDP ratios. Then, the money demand elasticity can be estimated using Moody's AAA corporate bond yields (from FRED) and the currency-to-GDP ratios, resulting in an estimated elasticity of -4.19.³⁰

Then, I jointly calibrate four parameters: the curvature of DM consumption σ , the fraction of currency transactions in the DM ρ , the value of government liabilities held by the public v , the fraction of bank assets that can be absconded δ , and the fixed cost of storing currency μ . The curvature parameter σ is calibrated to match the estimated money demand elasticity. Using currency-in-circulation data from FRED and the new M1 series from Lucas and Nicolini (2015), I calibrate the fraction of currency transactions in the DM ρ until the model generates the observed

²⁸The proportional cost of storing currency implies that the ELB on the nominal interest rate can be negative. However, the Federal Reserve might have faced legal and political issues with implementing negative nominal interest rates. As negative rates have not been explored in the U.S., it seems difficult to calibrate the proportional cost of storing currency with this model.

²⁹In the aftermath of the global financial crisis in 2007-2008, the demand for currency increased possibly due to non-transactional purposes. To calculate the elasticity of money demand specifically for transactions, I exclude post-crisis data, following the approach of Chiu, Davoodalhosseini, Jiang, and Zhu (2022) and Altermatt and Wang (2022).

³⁰The interest rate on liquid bonds (e.g., the 3-month Treasury Bill rate) may fluctuate due to changes in the liquidity premium. To exclude such a possibility, I consider the AAA corporate bond yield as the nominal interest rate on illiquid bonds and $\frac{\pi}{\beta} - 1$ as its theoretical counterpart.

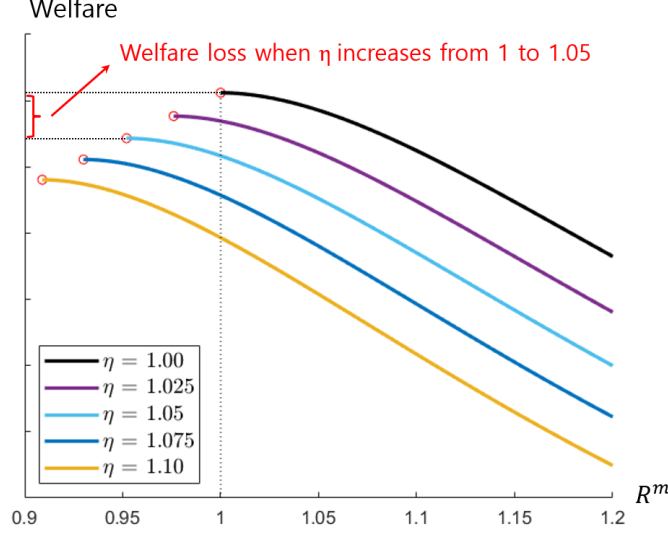


Figure 6: Monetary policy (R^m, η) and welfare

currency-to-M1 ratio. I use domestically-held public debt to GDP from FRED to calibrate the value of publicly-held government liabilities v .³¹ Another variable I use to calibrate parameters is the inflation rate. Along with other parameters, the fraction of assets that can be absconded δ is calibrated so that the model generates an inflation rate consistent with the observed rate of 1.06 percent. Finally, I set the fixed cost of storing currency μ to be 2 percent of cash payments.³²

A.2.2 Counterfactual Analysis

I consider three different scenarios where the fixed cost of theft κ is set to (i) 2.5 percent, (ii) 5 percent, and (iii) 10 percent of the current consumption level. With the calibrated parameters and each value of κ , I vary the non-par exchange rate η and compute the corresponding nominal interest rate on reserves that maximizes welfare, denoted by R_η^* . As depicted in Figure 6, an increase in η reduces the ELB on the nominal interest rate. However, the welfare level under the optimal nominal interest rate R_η^* decreases as η rises.

My approach to quantifying the welfare cost of increasing η involves measuring how much consumption private individuals would need to be compensated to tolerate the non-par exchange rate η . Suppose that, for any given value of η , the central bank sets the nominal interest rate at the optimal level, i.e., $R^m = R_\eta^*$. Then, the welfare measure can be expressed as:

$$\mathcal{W}(R^m = R_\eta^*, \eta) = \rho[u(x^c) - x^c] + (1 - \rho)[u(x^d) - x^d] - \alpha^b \kappa.$$

³¹I define domestically-held public debt as the total public debt net of public debt held by foreign and international investors.

³²Under this assumption, each buyer pays approximately 2 percent more to purchase goods in currency transactions, compensating for the seller's storage cost. While the fixed storage cost may indeed deviate from 2 percent, adjusting it within the range of 0 percent to 10 percent does not significantly affect the outcomes of the counterfactual analysis.

η	ELB	R_η^*	$(\Delta_\eta - 1) \times 100$		
			$\kappa = 2.5\%$	$\kappa = 5\%$	$\kappa = 10\%$
1.00	1.000	1.000	-	-	-
1.025	0.976	0.976	0.0561	0.1118	0.2236
1.05	0.952	0.952	0.1095	0.2183	0.4367
1.075	0.930	0.930	0.1604	0.3199	0.6399
1.10	0.909	0.909	0.2091	0.4168	0.8339

Table 4: ELB, optimal interest rate, and the welfare cost of reducing the ELB

If the consumption quantities in the DM are adjusted by a factor Δ , the adjusted level of welfare can be expressed as:

$$\mathcal{W}_\Delta(R^m = R_\eta^*, \eta) = \rho[u(\Delta x^c) - x^c] + (1 - \rho)[u(\Delta x^d) - x^d] - \alpha^b \kappa.$$

Then, I can compute the value Δ_η that solves $\mathcal{W}_{\Delta_\eta}(R^m = R_\eta^*, \eta > 1) = \mathcal{W}(R^m = R_\eta^*, \eta = 1)$. The welfare cost of introducing η can be measured as $\Delta_\eta - 1$ percent of consumption. If private individuals are compensated with this amount of consumption, they would be indifferent between the two policy choices: a one-to-one exchange rate and a non-par exchange rate.

Table 4 presents the ELB, the optimal nominal interest rate R_η^* , and the welfare cost of introducing a non-par exchange rate η for various fixed costs of theft κ . Notably, an increase in η reduces both the ELB and the optimal interest rate, regardless of the cost of theft. Recall that, with a fixed cost of storing currency close to zero ($\mu \approx 0$), the optimal monetary policy can be characterized by a modified Friedman rule ($\eta R^m \approx 1$). Thus, an increase in η leads to a decrease in the optimal nominal interest rate R_η^* . Assuming that monetary policy is conducted optimally for any given non-par exchange rate η , there would be no distortion in the equilibrium prices and allocations.

However, introducing a non-par exchange rate η increases the aggregate cost of theft in equilibrium. For instance, if the fixed cost of theft κ is 2.5 percent of the current consumption level, increasing η by 5 percent and 10 percent would cost, respectively, 0.11 percent and 0.21 percent of consumption. If κ is 5 percent of the current consumption level, the corresponding increases in η would cost 0.22 percent and 0.42 percent of consumption. Finally, with κ set at 10 percent of the current consumption level, the increases in η by 5 percent and 10 percent would cost 0.44 percent and 0.84 percent of consumption, respectively.³³

The welfare cost of introducing a non-par exchange rate can be better understood by comparing it with estimates for the welfare cost of another policy frequently discussed in the literature: the welfare cost of 10 percent inflation. Estimates for the welfare cost of 10 percent inflation are typically around 1 percent of consumption. Therefore, the welfare cost of introducing a non-par exchange rate seems significant.³⁴ It is important to note that the welfare cost of using a non-par exchange

³³Using different values for the fixed cost of storing currency μ does not significantly change the result. For example, if μ is 10 percent of cash payments and κ is 5 percent of the current consumption level, increasing η by 5 percent and 10 percent would cost, respectively, 0.2185 percent and 0.4172 percent of consumption.

³⁴For instance, the welfare cost of increasing inflation from 0 percent to 10 percent is reported as 0.62 percent of

rate critically depends on the fixed cost of investing in the theft technology κ . As κ increases, the welfare cost also increases proportionally.

A.3 Omitted Proofs

Proof of Lemma 1: First, consider the case where $\eta = 1$. If theft does not occur in equilibrium ($\alpha^b = 0$), sellers would be indifferent between depositing their currency with the central bank and engaging in side-trading with private banks ($\alpha^s \in [0, 1]$). For this no-theft equilibrium to be incentive-compatible, the cost of theft κ must be sufficiently high to satisfy $\kappa \geq \rho\alpha^s\eta c^s$. Moreover, if $\kappa > \rho\eta c^s$, no investment in theft ($\alpha^b = 0$) is optimal for buyers for any $\alpha^s \in [0, 1]$. If $\kappa = \rho\eta c^s$, buyers become indifferent between investing in theft and not investing, although $\alpha^b = 0$ can only be sustainable in equilibrium. Conversely, if $\kappa < \rho\eta c^s$, the fraction of sellers carrying currency into the TM must be sufficiently low to prevent theft. For κ to be sufficiently high to satisfy $\kappa \geq \rho\alpha^s\eta c^s$, the fraction α^s must fall within the range of $[0, \bar{\alpha}^s]$, where $\bar{\alpha}^s = \frac{\kappa}{\rho\eta c^s}$. Therefore, when $\eta = 1$, there exist a continuum of no-theft equilibria with $\alpha^b = 0$ and $\alpha^s \in [0, 1]$ for $\kappa \geq \rho\eta c^s$, and those with $\alpha^b = 0$ and $\alpha^s \in [0, \bar{\alpha}^s]$ for $\kappa < \rho\eta c^s$, where $\bar{\alpha}^s = \frac{\kappa}{\rho\eta c^s}$. However, a theft equilibrium does not exist in this case. This is because, when buyers steal currency ($\alpha^b \in (0, 1]$), no sellers would carry their currency into the TM ($\alpha^s = 0$), disincentivizing buyers from investing in the costly theft technology, making a theft equilibrium unsustainable.

Next, I consider the case where $\eta > 1$. If no theft occurs in equilibrium ($\alpha^b = 0$), sellers would prefer side-trading their currency with private banks over depositing it with the central bank ($\alpha^s = 1$). A necessary condition for this equilibrium to exist is $\kappa \geq \rho\eta c^s$. Therefore, a unique equilibrium exists with $\alpha^b = 0$ and $\alpha^s = 1$ for $\kappa \geq \rho\eta c^s$. If $\kappa < \rho\eta c^s$, buyers would choose to incur costs to acquire the theft technology ($\alpha^b = 1$). However, cases with $\alpha^b = 1$ (where buyers strictly prefer to invest in theft) cannot be supported in equilibrium because $\alpha^b = 1$ would lead to $\alpha^s = 0$, discouraging costly theft. Thus, in equilibrium, both buyers and sellers must be indifferent between their options. From (8) and (11), this implies:

$$\alpha^b = \frac{\eta - 1}{\eta}, \quad (68)$$

$$\alpha^s = \frac{\kappa}{\rho\eta c^s}. \quad (69)$$

Hence, there exists a unique equilibrium that satisfies (68) and (69) for $\kappa < \rho\eta c^s$.

Proof of Proposition 1: Note that, in equilibrium, equations (31) and (33) solve for x^c and x^d . For the comparative statics analysis with respect to R^m , I totally differentiate (31) and (33) and

consumption in Chiu and Molico (2010), 0.87 percent in Lucas (2000), and 1.32 percent (assuming that buyers make a take-it-or-leave-it offer to sellers) in Lagos and Wright (2005), among other studies.

evaluate the derivatives of x^c and x^d for $(R^m, \eta) = (1, 1)$ and $\mu = 0$, which yield:

$$\begin{aligned}\frac{dx^c}{dR^m} &= \frac{(1-\rho)[(1-\delta)u'(x) + \delta][(1-\delta)(1-\sigma)u'(x) + \delta]}{u''(x)[(1-\delta)(1-\sigma)u'(x) + \delta + \rho\beta\mu(1-\delta)u''(x)]} < 0, \\ \frac{dx^d}{dR^m} &= \frac{-\rho[(1-\delta)u'(x) + \delta][(1-\delta)(1-\sigma)u'(x) + \delta + \beta\mu(1-\delta)u''(x)]}{u''(x)[(1-\delta)(1-\sigma)u'(x) + \delta + \rho\beta\mu(1-\delta)u''(x)]} > 0,\end{aligned}$$

where $x^c = x^d = x$. It follows immediately that from (30) and (32), r^m , r^b , and π increase, while from the first argument in (35), the ELB remains unchanged.

Now, turn attention to the comparative statics analysis with respect to η . For convenience, let $\sigma = -\frac{xu''(x)}{u'(x)}$ so that $\sigma \in (0, 1)$. Then, evaluating the derivatives of x^c and x^d with respect to η for $(R^m, \eta) = (1, 1)$ and $\mu = 0$ yields:

$$\begin{aligned}\frac{dx^c}{d\eta} &= -\frac{\delta\{\rho\sigma u'(x) + (1-\rho)[(1-\delta)(1-\sigma)u'(x) + \delta][u'(x) - 1] - \beta\mu\rho u''(x)\}}{u''(x)[(1-\delta)(1-\sigma)u'(x) + \delta + \rho\beta\mu(1-\delta)u''(x)]} > 0, \\ \frac{dx^d}{d\eta} &= \frac{\rho\delta[(1-\sigma)u'(x) - 1 + \beta\mu u''(x)][(1-\delta)u'(x) + \delta]}{u''(x)[(1-\delta)(1-\sigma)u'(x) + \delta + \rho\beta\mu(1-\delta)u''(x)]}.\end{aligned}$$

Additionally, evaluating equation (33) for $(R^m, \eta) = (1, 1)$ and $\mu = 0$ yields:

$$\left[u'(x) + \frac{\delta}{1-\delta}\right](x + \rho\beta\mu) = v, \quad (70)$$

where x is increasing in v . The collateral constraint (70) does not bind in equilibrium if

$$v \geq \frac{x^* + \rho\beta\mu}{1-\delta}. \quad (71)$$

Let $\bar{v}$ denote the right-hand side of inequality (71), $\hat{x}$ denote the solution to $u'(x) = \frac{1}{1-\sigma}$, and $\hat{v}$ denote the solution to (70) when $x = \hat{x}$. Then, the derivatives of x^d with respect to η can be written as:

$$\begin{aligned}\frac{dx^d}{d\eta} &\leq 0, & \text{if } v \in (0, \hat{v}] \\ \frac{dx^d}{d\eta} &> 0, & \text{if } v \in (\hat{v}, \bar{v})\end{aligned}$$

Thus, from (30), an increase in η decreases r^m and r^b for $v \in (0, \hat{v}]$ and increases r^m and r^b for $v \in (\hat{v}, \bar{v})$. From (32) or

$$\pi = \beta R^m \left[u'(x^d) - \delta u'(x^d) + \delta \right],$$

an increase in η increases π for $v \in (0, \hat{v}]$ and decreases π for $v \in (\hat{v}, \bar{v})$. Finally, from the first argument in (35), the ELB falls.

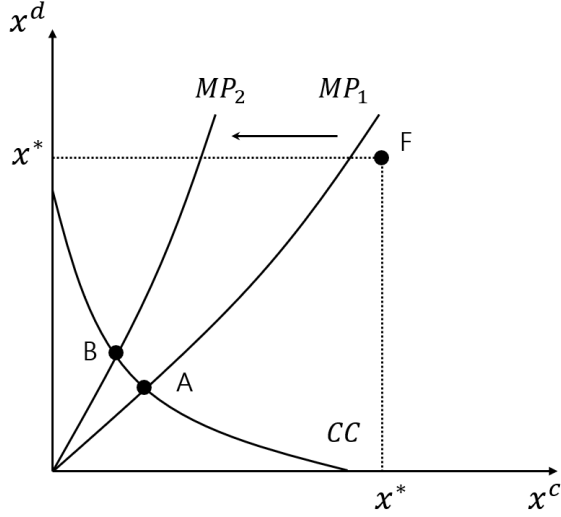


Figure 7: Effects of monetary policy (R^m, η)

Proof of Proposition 2: The DM consumption quantities, x^c and x^d , are determined by equations (37) and (39). Equation (39) can be expressed as:

$$F(x^c, x^d) = v, \quad (72)$$

and I can show that the function $F(\cdot, \cdot)$ is strictly increasing in both $0 \leq x^c < x^*$ and $0 \leq x^d < x^*$, as $-x \frac{u''(x)}{u'(x)} < 1$. Based on this property, equation (72) can be depicted by a downward-sloping locus in the (x^c, x^d) space, given v . Conversely, equation (37) can be depicted by an upward-sloping locus in the (x^c, x^d) space, given (R^m, η) .

For the comparative statics analysis, suppose the central bank raises R^m while holding η constant. As illustrated in Figure 7, the MP curve representing equation (37) shifts to the left, while the CC curve representing equation (72) remains unaffected. Thus, x^c decreases, and x^d increases. Then, from (36) and (38), π rises, and real interest rates (r^m, r^b) rise. From (27), (40), and (41), α^s increases, while α^b and the ELB remain unchanged.

Next, suppose that the central bank raises η while holding R^m constant. This monetary intervention also shifts the MP curve leftward, while leaving the CC curve unaffected. Consequently, x^c decreases, and x^d increases. Then, from (38), real interest rates (r^m, r^b) rise. Using (37), equation (36) can be written as:

$$\pi = \beta R^m \left[u'(x^d) - \delta u'(x^d) + \delta \right],$$

so π falls. From (27) and (40), α^b increases, and the ELB decreases. However, from (41), the effect on α^s is ambiguous since ηx^c can increase or decrease depending on parameters.

Proof of Proposition 3: The proof consists of two steps. First, I will identify monetary policies that maximize the welfare measure $\mathcal{W}$, assuming α^b is exogenously given. Then, I will determine

the optimal monetary policy considering α^b as endogenously determined in response to monetary policy.

In the first step, I solve the following maximization problem given $\alpha^b \in [0, 1]$:

$$\max_{(R^m, \eta)} \rho [u(x^c) - x^c] + (1 - \rho) \left[u(x^d) - x^d \right] - \alpha^b \kappa \quad (73)$$

subject to

$$\eta R^m = \frac{u'(x^c) - \delta u'(x^d) + \delta}{u'(x^d) - \delta u'(x^d) + \delta}, \quad (74)$$

$$\left[u'(x^c) + \frac{\delta}{1 - \delta} \right] \rho (x^c + \beta \mu) + \left[u'(x^d) + \frac{\delta}{1 - \delta} \right] (1 - \rho) x^d = v, \quad (75)$$

$$R^m \geq \frac{1}{\eta + \beta \gamma}, \quad \eta \geq 1 \quad (76)$$

Here, a monetary policy measure relevant to welfare in equilibrium is ηR^m , denoted as $\Omega \equiv \eta R^m$. Hence, from (76), Ω must satisfy $\Omega \geq \frac{\eta}{\eta + \beta \gamma}$. Differentiating the objective (73) with respect to Ω yields:

$$\frac{d\mathcal{W}}{d\Omega} = \rho [u'(x^c) - 1] \frac{dx^c}{d\Omega} + (1 - \rho) [u'(x^d) - 1] \frac{dx^d}{d\Omega}. \quad (77)$$

Let $\sigma = -\frac{xu''(x)}{u'(x)}$. Then, from total differentiation of (74) and (75), I derive:

$$\frac{dx^c}{d\Omega} = \frac{(1 - \rho) \left[(1 - \sigma)u'(x^d) + \frac{\delta}{1 - \delta} \right] [(1 - \delta)u'(x^d) + \delta]}{\Phi} < 0, \quad (78)$$

$$\frac{dx^d}{d\Omega} = \frac{-\rho \left[(1 - \sigma)u'(x^c) + \frac{\delta}{1 - \delta} + \beta \mu u''(x^c) \right] [(1 - \delta)u'(x^d) + \delta]}{\Phi} > 0, \quad (79)$$

where

$$\begin{aligned} \Phi = (1 - \rho)u''(x^c) \left[(1 - \sigma)u'(x^d) + \frac{\delta}{1 - \delta} \right] \\ + \rho u''(x^d) [(1 - \delta)\Omega + \delta] \left[(1 - \sigma)u'(x^c) + \frac{\delta}{1 - \delta} + \beta \mu u''(x^c) \right] < 0, \end{aligned}$$

for sufficiently low μ . A monetary policy Ω attains a local optimum if the resulting consumption quantities x^c and x^d lead to $\frac{d\mathcal{W}}{d\Omega} = 0$. From (77)-(79), I can characterize the optimal allocation x^c

and x^d as follows:

$$\begin{aligned}
\frac{d\mathcal{W}}{d\Omega} &= 0, \\
\Leftrightarrow [u'(x^c) - 1] \left[(1 - \sigma)u'(x^d) + \frac{\delta}{1 - \delta} \right] - [u'(x^d) - 1] \left[(1 - \sigma)u'(x^c) + \frac{\delta}{1 - \delta} + \beta\mu u''(x^c) \right] &= 0, \\
\Rightarrow [u'(x^c) - 1] \left[(1 - \sigma)u'(x^d) + \frac{\delta}{1 - \delta} \right] &\leq [u'(x^d) - 1] \left[(1 - \sigma)u'(x^c) + \frac{\delta}{1 - \delta} \right], \\
\Leftrightarrow \frac{u'(x^c) - 1}{(1 - \sigma)u'(x^c) + \frac{\delta}{1 - \delta}} &\leq \frac{u'(x^d) - 1}{(1 - \sigma)u'(x^d) + \frac{\delta}{1 - \delta}}.
\end{aligned}$$

Since the function $F(x) = \frac{u'(x) - 1}{(1 - \sigma)u'(x) + \frac{\delta}{1 - \delta}}$ is strictly decreasing in x , the above inequality is equivalent to $x^c \geq x^d$. Note that from (74) $x^c = x^d$ if $\Omega = 1$, and that as Ω rises, x^c decreases and x^d increases. Therefore, to achieve $x^c \geq x^d$, the optimal monetary policy Ω must satisfy $\Omega \leq 1$, which holds with equality if and only if $\mu = 0$.

In the first step, I have shown that an optimal monetary policy must be a combination of (R^m, η) such that $\eta R^m \leq 1$. All optimal combinations of (R^m, η) lead to the same gains from trade in the DM, $\rho[u(x^c) - x^c] + (1 - \rho)[u(x^d) - x^d]$. However, as the non-par exchange rate η rises, the fraction of buyers who choose to steal currency in the TM α^b increases ((40)), leading to a increase in the total cost of theft in the TM. Therefore, the optimal monetary policy is given by $\eta = 1$ and $R^m \leq 1$.

Even if the optimal R^m is constrained by (76), the optimal interest rate is the lower bound $\frac{1}{\eta + \beta\gamma}$. An increase in η allows R^m to decrease further, but it acts to increase the current level of Ω . So, an increase in η cannot increase the sum of surpluses from trade in the CM and the DM, represented by the first two terms in the welfare measure (73), while increasing the total theft cost in the TM.

Proof of Proposition 4: Suppose that the cost of theft κ is sufficiently high to prevent theft ($\alpha^b = 0$) in equilibrium. To show that social welfare increases with η , consider the scenario where the central bank sets the nominal interest rate on reserves R^m to achieve $x^c = x^d = x$ given a non-par exchange rate η . While this policy may not be optimal, it helps understand the optimal level of η . When $x^c = x^d = x$, equations (31) and (33) can be rewritten as:

$$R^m = R^b = \frac{\eta u'(x) - \delta u'(x) + \delta}{\eta [u'(x) - \delta u'(x) + \delta]}, \quad (80)$$

$$u'(x)[x + \rho\beta\mu] + \frac{[\rho + (1 - \rho)\eta]\delta x + \rho\delta\beta\mu}{(1 - \delta)\eta} = v. \quad (81)$$

In this case, equation (81) solves for x , and then equation (80) solves for R^m . If the value of the consolidated government debt v is sufficiently low, or

$$v \leq \frac{[(1 - \delta\rho)\eta + \delta\rho]x^* + \rho\beta\mu[(1 - \delta)\eta + \delta]}{(1 - \delta)\eta},$$

then x increases with η for a sufficiently low μ . Since the level of welfare can be written as $\mathcal{W} = u(x) - x$ in this equilibrium, an increase in η effectively increases welfare as long as the nominal interest rate R^m can be adjusted to achieve $x^c = x^d = x$. However, according to Proposition 1, an increase in η must be accompanied by an increase in R^m to maintain the same consumption quantities across two types of DM transactions, implying that the choice of R^m is not constrained by the ELB. Although the optimal R^m may not satisfy $x^c = x^d$, the fact that social welfare increases with η remains unchanged. Finally, η must be sufficiently low to prevent buyers from investing in the theft technology. Therefore, at the optimum, η is chosen such that buyers are indifferent between stealing currency and not stealing.

Suppose further that there is no fixed cost of holding currency at the beginning of the TM, i.e., $\mu = 0$. Consider the following maximization problem given $\eta \geq 1$:

$$\max_{(R^m, \eta)} \rho [u(x^c) - x^c] + (1 - \rho) [u(x^d) - x^d] \quad (82)$$

subject to

$$\eta R^m = \frac{\eta u'(x^c) - \delta u'(x^d) + \delta}{u'(x^d) - \delta u'(x^d) + \delta}, \quad (83)$$

$$\left[u'(x^c) + \frac{\delta}{(1 - \delta)\eta} \right] \rho x^c + \left[u'(x^d) + \frac{\delta}{1 - \delta} \right] (1 - \rho)x^d = v, \quad (84)$$

$$R^m \geq \max \left\{ \frac{1}{\eta + \beta\gamma}, \frac{\eta}{(\eta + \beta\gamma)[(1 - \delta)u'(x^d) + \delta]} \right\}. \quad (85)$$

Differentiating the objective (82) with respect to R^m yields:

$$\frac{d\mathcal{W}}{dR^m} = \rho [u'(x^c) - 1] \frac{dx^c}{dR^m} + (1 - \rho) [u'(x^d) - 1] \frac{dx^d}{dR^m}. \quad (86)$$

Letting $\sigma = -\frac{xu''(x)}{u'(x)}$ and totally differentiating (83) and (84) gives:

$$\begin{aligned} \frac{dx^c}{dR^m} &= \frac{\eta(1 - \rho) \left[(1 - \sigma)u'(x^d) + \frac{\delta}{1 - \delta} \right] [(1 - \delta)u'(x^d) + \delta]}{\Lambda} < 0, \\ \frac{dx^d}{dR^m} &= \frac{-\eta\rho \left[(1 - \sigma)u'(x^c) + \frac{\delta}{(1 - \delta)\eta} \right] [(1 - \delta)u'(x^d) + \delta]}{\Lambda} > 0, \end{aligned}$$

where

$$\begin{aligned} \Lambda &= (1 - \rho)\eta u''(x^c) \left[(1 - \sigma)u'(x^d) + \frac{\delta}{1 - \delta} \right] \\ &\quad + \rho u''(x^d) \left[(1 - \sigma)u'(x^c) + \frac{\delta}{(1 - \delta)\eta} \right] [\eta R^m(1 - \delta) + \delta] < 0. \end{aligned}$$

Then, I can evaluate the derivative of $\mathcal{W}$, or equation (86), for $\eta R^m = 1$. Noting that $\eta u'(x^c) = u'(x^d)$ from (83), I obtain

$$\left. \frac{d\mathcal{W}}{dR^m} \right|_{\eta R^m=1} = \frac{\eta(1-\eta)\rho(1-\rho) [(1-\delta)u'(x^d) + \delta]}{\eta^2(1-\rho)u''(x^c) + \rho u''(x^d)} \geq 0, \quad (87)$$

implying that $\eta R^m \geq 1$ at an optimum.

Next, differentiating the objective (82) with respect to η gives:

$$\frac{d\mathcal{W}}{d\eta} = \rho [u'(x^c) - 1] \frac{dx^c}{d\eta} + (1-\rho) [u'(x^d) - 1] \frac{dx^d}{d\eta}. \quad (88)$$

I totally differentiate (83) and (84) to derive $\frac{dx^c}{d\eta}$ and $\frac{dx^d}{d\eta}$, and then evaluate the derivatives for $\eta R^m = 1$, which yields:

$$\begin{aligned} \frac{dx^c}{d\eta} &= \frac{\delta \rho x^c u''(x^d) - \delta \eta (1-\rho) [u'(x^d) - 1] [(1-\delta)(1-\sigma)u'(x^d) + \delta]}{\eta [(1-\delta)(1-\sigma)u'(x^d) + \delta] [\rho u''(x^d) + \eta^2(1-\rho)u''(x^c)]} > 0, \\ \frac{dx^d}{d\eta} &= \frac{\delta \rho \{ \eta x^c u''(x^c) + [u'(x^d) - 1] [(1-\delta)(1-\sigma)u'(x^d) + \delta] \}}{\eta [(1-\delta)(1-\sigma)u'(x^d) + \delta] [\rho u''(x^d) + \eta^2(1-\rho)u''(x^c)]}. \end{aligned}$$

Substituting the above expressions into (88) gives:

$$\left. \frac{d\mathcal{W}}{d\eta} \right|_{\eta R^m=1} = \frac{\Gamma + \rho \delta x^c \{ \rho u''(x^d) [u'(x^c) - 1] + \eta (1-\rho) u''(x^c) [u'(x^d) - 1] \}}{\eta [(1-\delta)(1-\sigma)u'(x^d) + \delta] [\rho u''(x^d) + \eta^2(1-\rho)u''(x^c)]}, \quad (89)$$

where

$$\Gamma = \rho \delta (1-\rho)(\eta-1) [u'(x^d) - 1] [(1-\delta)(1-\sigma)u'(x^d) + \delta] \geq 0.$$

From (89), the derivative of $\mathcal{W}$ is strictly positive if $\eta = R^m = 1$, i.e., $\left. \frac{d\mathcal{W}}{d\eta} \right|_{\eta=R^m=1} > 0$. This implies that the monetary policy at $\eta = R^m = 1$ is not optimal. Therefore, from (87) and (89), I conclude that the optimal monetary policy is characterized by $\eta R^m > 1$.

Proof of Proposition 5: Suppose that $\theta = 0$ in equilibrium. From equations (17)-(19) and (46),

$$\eta R^m [u'(x^d) - \delta u'(x^d) + \delta] = u'(x^o), \quad (90)$$

$$u'(x^o) = u'(x^c) - \delta u'(x^d) + \delta, \quad (91)$$

where (x^c, x^d) are the off-equilibrium consumption quantities in DM transactions, if a buyer were to participate in banking contracts. It can be shown that $\left| \frac{d[u'(x^d) - \delta u'(x^d) + \delta]}{d[\eta R^m]} \right| < 1$, so from (90) x^o increases with a decrease in ηR^m . However, the limited quantity of collateral (represented by $v < x^* + \beta \mu$) implies that, from (60), the highest possible quantity for x^o is $\bar{x}$ that solves

$(\bar{x} + \beta\mu)u'(\bar{x}) = v$, and $\bar{x} < x^*$. So, any ηR^m that leads to x^o higher than $\bar{x}$ cannot be supported in equilibrium, implying that, from (90),

$$R^m \geq \frac{u'(\bar{x})}{\eta[u'(\underline{x}^d) - \delta u'(\underline{x}^d) + \delta]}, \quad (92)$$

where $\underline{x}^d$ is the off-equilibrium consumption quantity in DM transactions involving bank claims that is consistent with (91). Also, any R^m higher than the lower bound in (92) results in x^o such that $(x^o + \beta\mu)u'(x^o) < v$. This implies that $0 < \theta \leq 1$ and $U^b \geq U^o$ in equilibrium. So, by continuity, the off-equilibrium consumption quantities (x^c, x^d) when $R^m = \frac{u'(\bar{x})}{\eta[u'(\underline{x}^d) - \delta u'(\underline{x}^d) + \delta]}$ must satisfy (91) and $U^b = U^o$ from (48)-(49) given $x^o = \bar{x}$.

Recall that the nominal interest rate R^m must also satisfy (43), the no-arbitrage condition for agents from holding currency across periods. To prove that the lower bound on R^m given in (92) is always higher than the lower bound in (43), I claim that the following condition must hold in equilibrium if the fixed cost of holding currency μ is close to zero:

$$\frac{u'(\bar{x})}{u'(\underline{x}^d) - \delta u'(\underline{x}^d) + \delta} \geq 1. \quad (93)$$

To prove this claim, suppose $\eta R^m = 1$. In this case, deposit contracts allow buyers to consume the same quantity of goods across two types of DM transactions, i.e., $x^c = x^d = x$. Then, the quantity of DM consumption for buyers opting out of deposit contracts is higher than the quantity for those holding deposit contracts ($x^o > x$) because, from (90)-(91),

$$u'(x^o) = (1 - \delta)u'(x) + \delta.$$

This implies that the expected utility for buyers opting out of contracts is higher than the expected utility for those opting in, because from (48)-(49),

$$U^o - U^b = [u(x^o) - x^o u'(x^o)] - [u(x) - x u'(x)] - \beta\mu [u'(x^o) - \rho u'(x)] > 0,$$

for a sufficiently low μ . Thus, the policy combination of $\eta R^m = 1$ leads to complete disintermediation ($\theta = 0$), which implies that the lower bound on the policy combination, $\frac{u'(\bar{x})}{u'(\underline{x}^d) - \delta u'(\underline{x}^d) + \delta}$, must be higher than or equal to one. Also, $\frac{u'(\bar{x})}{\eta[u'(\underline{x}^d) - \delta u'(\underline{x}^d) + \delta]} > \frac{1}{\eta + \beta\gamma}$ for any $\eta \geq 1$ because, as η rises, $\eta + \beta\gamma$ increases more than $\eta[u'(\underline{x}^d) - \delta u'(\underline{x}^d) + \delta]$. Therefore, the ELB on the nominal interest rate is always determined by (92).

Proof of Proposition 6: First, note that an equilibrium with $\theta = 0$ exists only when R^m is set at the ELB defined in (58). As mentioned in the Proof of Proposition 5, any R^m higher than the ELB implies $x^o < \bar{x}$, where $\bar{x}u'(\bar{x}) = v$. Since $x^o u'(x^o) < v$ and the collateral constraint must bind in an equilibrium where v is sufficiently low, there must exist some buyers participating in banking contracts, i.e., $\theta > 0$. Thus, for any R^m higher than the ELB, $\theta > 0$ in equilibrium.

Next, consider an equilibrium with $0 < \theta < 1$. In this case, (x^c, x^d, x^o, θ) must satisfy:

$$\eta R^m = \frac{u'(x^c) - \delta u'(x^d) + \delta}{u'(x^d) - \delta u'(x^d) + \delta}, \quad (94)$$

$$(1 - \rho)\theta x^d \left[u'(x^d) + \frac{\delta}{1 - \delta} \right] + \rho\theta x^c \left[u'(x^c) + \frac{\delta}{1 - \delta} \right] + (1 - \theta)x^o u'(x^o) = v, \quad (95)$$

$$u'(x^o) = u'(x^c) - \delta u'(x^d) + \delta, \quad (96)$$

$$\rho [u(x^c) - x^c u'(x^c)] + (1 - \rho) [u(x^d) - x^d u'(x^d)] = u(x^o) - x^o u'(x^o). \quad (97)$$

Suppose that there is an increase in ηR^m . Then, from (94), x^c decreases while x^d increases. From (96), x^o decreases with the decrease in x^c and the increase in x^d . Also, from (58) and (96), a necessary condition for this equilibrium to exist is $x^c < x^d$, implying that U^b decreases as x^c falls and x^d rises. As the left-hand side of (97) (representing U^b) decreases, x^o must fall in equilibrium, consistent with (96). Then, from (95), θ must rise in equilibrium. The effects of an increase in ηR^m on $(\pi, \alpha^b, \alpha^s)$ can be straightforwardly derived from (61), (64), and (65).

Given that an increase in ηR^m leads to a decrease in x^c and x^o and an increase in x^d and θ , there exists a ηR^m , denoted by $\bar{\Omega}$, that satisfies equation (94), with x^c and x^d that consist of the solution to equations (95)-(97) when $\theta = 1$. Therefore, I can conclude that, in equilibrium, $0 \leq \theta < 1$ if $\frac{u'(\bar{x})}{u'(\underline{x}^d) - \delta u'(\underline{x}^d) + \delta} \leq \eta R^m < \bar{\Omega}$ and $\theta = 1$ if $\eta R^m \geq \bar{\Omega}$, where $\bar{x}$ and $\underline{x}^d$ are the quantities defined in the Proof of Proposition 5.

Proof of Proposition 7: In the Proof of Proposition 6, I have shown that, for any $\eta R^m > \frac{u'(\bar{x})}{u'(\underline{x}^d) - \delta u'(\underline{x}^d) + \delta}$, the fraction θ is positive and $x^c < x^d$ in equilibrium. This implies that both the left-hand and the right-hand side of (97) increase (i.e., both U^b and U^o increase) as x^c and x^o rise and x^d falls. So, given η , a decrease in R^m increases the welfare measure $\mathcal{W}$ by increasing x^c and x^o and decreasing x^d and θ . Then, by continuity, the maximum $\mathcal{W}$ can be obtained when $\eta R^m = \frac{u'(\bar{x})}{u'(\underline{x}^d) - \delta u'(\underline{x}^d) + \delta}$ given η . However, if the central bank conducts monetary policy (R^m, η) such that $\eta R^m = \frac{u'(\bar{x})}{u'(\underline{x}^d) - \delta u'(\underline{x}^d) + \delta}$, a higher η only implies a higher α^b without increasing the sum of surpluses from trade in the CM and the DM. As a higher α^b leads to a larger total cost of theft, the welfare measure $\mathcal{W}$ is maximized if and only if $\eta = 1$ and $R^m = \frac{u'(\bar{x})}{u'(\underline{x}^d) - \delta u'(\underline{x}^d) + \delta}$.